

When Shocks Divide: Religiosity and Science in the Time of Adversity*

Enrico Berkes Davide M. Coluccia Gaia Dossi Mara P. Squicciarini

This Version: May, 2026

Abstract

We show that U.S. counties hit harder by the 1918 influenza pandemic became both more religious and more science-oriented, a result at odds with the existing literature that documents a negative relationship between religiosity and science. Using a newly assembled name-based measure of religiosity, we find that this coexistence was driven by heterogeneous responses: individuals from more religious backgrounds became more religious, while those from less religious backgrounds were more likely to turn toward science. As a result, the pandemic widened preexisting differences in religiosity and increased polarization within society.

JEL: J24, N13, Z12

Keywords: Religiosity, Science, Innovation, Influenza Pandemic

*We are grateful to Ran Abramitzky, Philipp Ager, Ainoa Aparicio Fenoll (discussant), Sascha Becker, Roland Bénabou, Davide Cantoni, Martin Fiszbein, Roberto Galbiati, Claudia Goldin, Reka Juhasz, Ben Marx, Joel Mokyr, Nathan Nunn, Thomas Piketty, Jared Rubin, Marco Tabellini, Felipe Valencia, John Van Reenen, Nico Voigtländer, Bruce Weinberg, Noam Yuchtman, and Katia Zhuravskaya for comments and discussions. We thank seminar audiences at the 2022 ASREC Annual Conference, the ASREC Workshop on the Economics of Religion, Barcelona, Bergen, Bicocca, Bocconi, Bolzano, Boston University, Brown, Geneva, Harvard, Louvain la Neuve, LSE, LUISS, Mannheim, the 6th Marco Fanno Alumni Workshop, Northwestern, Parthenope, Pavia, Pompeu Fabra, PSE/Sciences Po PEPES, Stanford, UC Berkeley, the University of British Columbia, Utrecht, and Warwick for feedback and suggestions. We also thank Luca Favero, Lorenzo Pedretti, Luisa Pomarici, and Sara Veronesi for excellent assistance during the construction of the dataset, and Ran Abramitzky, Andy Ferrara, and Marco Tabellini for kindly sharing data. Berkes: University of Maryland, Baltimore County, enrico.berkes@umbc.edu; Coluccia: University of Bristol, davide.coluccia@bristol.ac.uk; Dossi: Einaudi Institute for Economics and Finance, gaia.dossi@eief.it; Squicciarini: Bocconi University and CEPR, mara.squicciarini@unibocconi.it.

1 Introduction

Throughout history, societies have experienced shocks that disrupt economic, social, and cultural life. Pandemics, natural disasters, and other crises impose sudden and severe hardships, raising a central question: How do individuals and communities respond to adverse events? A large literature in economics and related fields has examined this question from different angles, documenting changes in political preferences, migration, and social norms. A less explored question, however, is whether these changes bring societies together or drive them apart.

In this paper, we focus on two factors that have been shown to respond to adversity, namely religion and science, in the context of the 1918 influenza pandemic in the United States.¹ Our difference-in-differences analysis at the county-level shows that societies can become *both* more religious and more science-oriented at the same time, a finding that appears in contrast with a large body of work documenting a negative relationship between religiosity and science (e.g., Bénabou, Ticchi, and Vindigni, 2015, 2022; Lecce, Ogliari, and Squicciarini, 2021). To explore the mechanism behind this pattern, we study how individuals *within* counties reacted to the shock. We uncover heterogeneous responses, with religion and science acting as alternative ways through which different individuals react to adversity. As a result, the pandemic widened pre-existing differences in religiosity and increased polarization within society.²

Historical records on the Great Influenza Pandemic indicate that many individuals strengthened their religious faith to cope with the pandemic. At the same time, the period following the pandemic witnessed increased scientific progress and major medical advances.³ To conduct our empirical analysis, we construct a novel, data-driven measure of religiosity. This measure is based on the naming patterns of babies born between 1900 and 1930 from the historical full-count censuses. Complementing these data with information from the Census of Religious Bodies, we empirically identify religious names and construct a measure of “revealed religiosity.” The underlying idea is that a child’s first name conveys information about their parents’ religiosity.⁴ We measure scientific progress through two related but distinct variables: the share of the population employed in science, technology, engineering,

¹For example, Bentzen (2019) documents that, across countries and within regions, individuals become more religious when hit by earthquakes. Moscona (2022) finds an increase in innovation efforts towards technologies that mitigate environmental distress in U.S. counties more exposed to the Dust Bowl during the 1930s.

²These findings reconcile our aggregate results with the existing literature on the negative relationship between religiosity and science.

³An increase in religiosity and scientific progress has also been documented after the COVID-19 outbreak. Bentzen (2021), using Google search data, finds a sharp increase in the intensity of prayers during the early days of the pandemic. Agarwal and Gaule (2022) show that the COVID-19 pandemic catalyzed R&D expenditure on pharmaceuticals and digital technologies.

⁴We validate the name-based religiosity measure using alternative indicators, including the share of biblical or saint names and the share of individuals affiliated with a religious denomination.

and mathematics (STEM) occupations and the number of patents granted in each county. These measures capture both human capital investments in science and the production of new ideas.⁵

Using a difference-in-differences framework, we first show that counties hit harder by the pandemic experienced an increase in religiosity. Moving from a county at the 25th percentile of the excess deaths distribution—our main measure of the intensity of the influenza shock—to one at the 75th percentile led to a 10% increase in overall religiosity in that county. Additionally, employment in scientific occupations and patenting grew disproportionately more in counties hit harder by the pandemic. Event-study analyses indicate the absence of pre-pandemic differences between treated and control counties, supporting the validity of our research design.

These results hold under a variety of alternative specifications, samples, and robustness checks. We show that our results are not driven by migration, small counties, or specific states, and that they remain robust when weighting regressions by population or running the analysis at the city level.⁶ For patents, we show that the increase in innovation was not driven by low-importance patents. We also rule out potential confounding shocks such as World War I and document that counties hit harder by the pandemic were not on systematically different trajectories before 1918.⁷

These aggregate responses could arise either because individuals became both more religious and more science-oriented, or because different people reacted in opposite ways, with some turning toward religion and others toward science. To investigate these potential mechanisms, we study how individuals within the same county responded to the pandemic and obtain three main results. First, the shock increased the gap in religiosity between individuals in STEM occupations and the rest of the population. Then, we focus on people of different religious backgrounds.⁸ Individuals from more religious backgrounds became more religious, while those from less religious backgrounds were more likely to enter scientific occupations. As a result, the pandemic widened the overall distribution of religiosity, with areas more exposed to the shock becoming more polarized: highly religious and less religious individuals moved further apart along this dimension.

One key question is why some individuals became more religious while others selected a scientific occupation. Our findings on religiosity are in line with the religious coping hypothesis, which

⁵In the rest of the paper, we refer to science and scientific progress interchangeably. We also use the term “scientific attitudes” to encompass both occupational choice (STEM) and innovation activity (patents).

⁶To show that results are not driven by migration, we first run a placebo exercise testing the impact of the pandemic on the names of adults. The results show no effect of the shock on adults’ names, which we interpret as evidence that the observed increase in religiosity was not driven by ex-ante more religious people moving to areas hit harder by the shock. We then show that our results hold when excluding from the estimation sample all individuals who, in the 1930 census, reside in a state different from the one where they were born.

⁷For a systematic overview of alternative mechanisms and the corresponding robustness checks, see the summary table in Appendix C.

⁸We measure religious background using individuals’ own names (rather than their children’s) to capture a person’s religious upbringing rather than their current faith.

posits that religious faith can represent a coping device to deal with personal distress following a negative shock.⁹ What motivated people to turn to science is less obvious. Importantly, though, our evidence suggests that this pattern reflects differences in individual responses rather than labor market forces alone. Scientific occupations expanded more in areas hit harder by the pandemic, consistent with increased demand for science and technology. However, the individual-level analysis reveals that this expansion was driven disproportionately by individuals from less religious backgrounds, while individuals from more religious backgrounds became more religious. These results also hold when restricting the sample to individuals in skilled occupations, suggesting that religious background is not merely proxying for human capital or socio-economic status.

The individual-level evidence reconciles the county-level findings with the existing literature. Although counties became both more religious and more innovative, individuals reacted differently to the same shock depending on their scientific orientation or religious background. Religion and science thus emerged as alternative ways of reacting to the pandemic. Consequently, the gap in religiosity between science-oriented and religious individuals widened relative to the pre-pandemic period, thus increasing the polarization of religious sentiment within society.

Related Literature. This paper is closely related to the literature studying how societies react to negative shocks. Previous work shows that, in accordance with the religious coping hypothesis (Pargament, 2001; Ano and Vasconcelles, 2005; Norenzayan and Hansen, 2006), natural disasters are associated with an increase in religiosity, both historically (e.g. Belloc, Drago, and Galbiati, 2016; Bentzen, 2019) and in contemporary scenarios (Sibley and Bulbulia, 2012; Bentzen, 2021).¹⁰ Another set of studies documents that economic crises (Babina, Bernstein, and Mezzanotti, 2023), wars (Gross and Sampat, 2021), climate change (Miao and Popp, 2014; Moscona, 2022; Clemens and Rogers, 2023), and pandemics (Gross and Sampat, 2021; Agarwal and Gaule, 2022) all influence innovation activity. To the best of our knowledge, this is the first study to provide evidence that natural disasters may foster a contemporaneous increase in religiosity *and* innovation, and also the first to document the ensuing polarization of religiosity within society.¹¹

Additionally, we speak to the literature on the drivers of polarization in society. Most of the existing work in economics focuses on the causes of *political* polarization. Previous studies have documented the role of economic shocks (e.g., Autor, Dorn, Hanson, and Majlesi, 2020), media (Gentzkow and

⁹An alternative explanation could be that individuals turn to religion as an insurance mechanism against the negative economic effects of the pandemic. See discussion in Section 6 for details.

¹⁰The religious coping hypothesis, first developed in the psychology literature, posits that people who are subject to economic and social shocks turn to religious faith as a coping device to deal with personal distress.

¹¹Many studies have looked at the impact of natural disasters on, among others, social norms (e.g. Posch, 2022), migration (e.g. Boustan, Kahn, and Rhode, 2012), and economic activity (e.g. Boustan, Kahn, Rhode, and Yanguas, 2020), while other works have focused on how negative shocks may affect the direction of innovation (e.g. Hanlon, 2015).

Shapiro, 2011; Martin and Yurukoglu, 2017), and ethnic fragmentation (Montalvo and Reynal-Querol, 2005; Esteban, Mayoral, and Ray, 2012) as drivers of political polarization. To the best of our knowledge, we are the first to document that a negative shock can lead to a polarization of religious beliefs.

Third, this study contributes to the broad literature on the economics of religion, pioneered by Weber (1905). In particular, we contribute to those studies that analyze the link between religiosity and science.¹² While most papers adopt a historical (Deming, 2010; Mokyr, 2011), theoretical (Bénabou et al., 2022), or cross-sectional perspective (Bénabou et al., 2015, 2022), to our knowledge, we are the first to study the interaction between religion and science in a panel setting and to uncover the individual-level dynamics behind their coevolution.¹³

Finally, we add to a growing literature using informational content of names to measure individuals' characteristics. Names have been used, for example, to measure race and ethnicity (Abramitzky, Boustan, and Eriksson, 2016; Fouka, 2019), individualism (Bazzi, Fiszbein, and Gebresilasse, 2020), socioeconomic background (Biavaschi, Giulietti, and Siddique, 2017; Olivetti, Paserman, Salisbury, and Weber, 2020), nationalism (Jurajda and Kovač, 2021), and religiosity (Abramitzky et al., 2016; Andersen and Bentzen, 2022). While all of these papers assume a preexisting rule to classify names (e.g., whether one has a biblical or saint name or a name shared by a major religious figure), to the best of our knowledge, we are the first to identify the religiosity of names *directly from the data*.¹⁴

The rest of the paper is structured as follows. Section 2 describes the Great Influenza Pandemic in the United States and discusses the historical evidence on its effects on religiosity and innovation. Section 3 presents the data and our indicator of religiosity. Section 4 discusses the empirical strategy and the core results. Section 5 explores the possible mechanisms underlying our findings. Section 6 discusses the results and the limitations of the analysis. Section 7 concludes.

2 Historical Background

This section provides an overview of the Great Influenza Pandemic in the United States and its influence on religion and scientific progress.

¹²Other studies analyze the relationship between religion and accumulation of human capital, more broadly (Becker and Woessmann, 2009; Botticini and Eckstein, 2012; Squicciarini, 2020). For an overview of the literature on the economics of religion, see Iannaccone (1998), Iyer (2016), and Becker, Rubin, and Woessmann (2021).

¹³Lecce et al. (2021) study how religiosity impacts the birth and migration of scientists in 19th-century French cantons, and Andersen and Bentzen (2022) show that individuals with religious names are less likely to become engineers, scientists, or doctors, and that cities with more religious individuals grew more slowly. However, none of these studies analyzes how an adverse shock affects society's heterogeneous response in terms of religion and science, and the underlying individual-level dynamics.

¹⁴For details on how we construct our religiosity measure, see Section 3.

2.1 The Great Influenza Pandemic

Between 1918 and 1919, the Great Influenza Pandemic—also known as “Spanish Flu”¹⁵—led to the death of an estimated 40 million people worldwide (approximately 1 in 30 people). It was one of the deadliest natural disasters in modern times (Barro, Ursúa, and Weng, 2020). In the United States, the pandemic started in the spring of 1918 with sporadic outbreaks. Then, a second, more severe wave began in September 1918. The final wave started in January 1919, ending that spring. In total, it killed about 500,000 Americans, corresponding to 0.7% of the U.S. population (Crosby, 1989).¹⁶

Historical and modern accounts suggest that the pandemic hit across the U.S. quasi-randomly. Crosby (1989) writes that the states with the highest mortality displayed diverse geographical, climatic, and demographic characteristics. The pandemic hit with varying intensity within states as well. For example, in Minnesota, the death rate in Saint Paul was about 70% higher than in Minneapolis, despite the two cities being just 8 miles apart. In Ohio, Dayton experienced an 80% higher mortality rate than Columbus, even though the two cities had similar demographic characteristics (Huntington, 1923; Almond, 2006).

The infection was caused by A/H1N1 influenza virus strains, whose origin remains unknown. Neither antiviral drugs to treat the primary disease nor antibiotics to cure secondary bacterial infections were available. Doctors had to rely on an array of mostly ineffective—sometimes fatal—medicines such as aspirin and quinine (Spinney, 2018). It is debated whether nonpharmaceutical interventions (NPIs)—such as wearing masks, canceling public events, closing schools, and implementing isolation and quarantine measures—were effective in limiting the spread of the disease.¹⁷

2.2 The Pandemic and Religion

A large literature documents that individuals become more religious in response to adverse events. One explanation comes from the “religious coping hypothesis,” which posits that individuals turn to religious beliefs or practices as a way to cope with sudden, dramatic circumstances (Pargament, 2001).¹⁸

¹⁵The Great Influenza Pandemic is popularly known as “Spanish Flu” because media in Spain—which was neutral during World War I (WWI)—were free to report news on this disease. Conversely, countries involved in WWI imposed press censorship on the topic. This gave the (incorrect) impression that Spain was either more severely hit by the disease or that the pandemic had originated in Spain.

¹⁶By comparison, COVID-19 caused 1.13 million deaths in the United States, approximately 0.3% of the U.S. population, between March 2020 and February 2023 (<https://covid.cdc.gov/covid-data-tracker/#datatracker-home>; accessed February 12, 2023).

¹⁷Some authors assert that NPIs were effective in reducing mortality (e.g., Markel, Lipman, Navarro, Sloan, Michalsen, Stern, and Cetron, 2007; Berkes, Deschenes, Gaetani, Lin, and Severen, 2023), while others show that the effect of NPIs on overall deaths was small and statistically insignificant (e.g., Barro, 2022).

¹⁸For example, Bentzen (2019) documents that individuals become more religious when hit by earthquakes. Religion may also represent an insurance mechanism when negative shocks occur. Ager, Hansen, and Lønstrup (2016) shows that after the 1927 Great Mississippi Flood, demand for social insurance led to higher churchgoing, while Ager and Ciccone

The influenza pandemic inflicted substantial emotional and socioeconomic distress and could have acted as a powerful amplifier of religious sentiments (Phillips, 2020). Historical records document that spiritualism gained momentum in the aftermath of the pandemic. For instance, in the United States, modern evangelism benefited from the pandemic, as evidenced by a sharp rise in the circulation of evangelical magazines (Frost, 2020). Membership in Christian Science also soared during these years, reaching an all-time peak in the 1930s.¹⁹ Catholics and Orthodox Jews identified the influenza as a manifestation of divine anger, the expiation of which called for prayers.²⁰

2.3 The Pandemic and Science

Historical evidence suggests that the period after 1918 was one of sharp intellectual and scientific progress and that the Great Influenza Pandemic was particularly influential in shaping the development of medical sciences (Barry, 2020). Despite being ineffective during the pandemic, medicine evolved enormously in subsequent years. In 1928, Alexander Fleming discovered the medical use of penicillin in treating bacterial infections. By the 1930s, virology had become an established branch of medicine, and the first influenza vaccines were being developed (Spinney, 2018). During this time, medicine became more “scientific” and, hence, effective (Barry, 2020).

These advancements in medicine went hand in hand with increased trust in scientific progress. For instance, in her personal journal, Canadian author L. M. Montgomery wrote, “[...] the Spirit of God no longer works through the church for humanity. It did once but it has worn out its instrument and dropped it. Today it is working through Science” (Montgomery, 1924, p. 211). Barry (2020) argues that the pandemic was the key driver behind this paradigm shift because it fostered scientific thinking in the face of such a sudden and dramatic shock.

This overview suggests that the 1918–1919 pandemic fostered both scientific progress and religiosity—a result that might seem at odds with theoretical and empirical evidence, which depicts religion and science as opposing forces (e.g., Bénabou et al., 2015, 2022; Lecce et al., 2021). In this paper, we provide evidence that the influenza shock led to a simultaneous increase in religiosity and scientific progress at the aggregate level. We then reconcile this apparent puzzle by showing that it induced polarization within society, with some people turning to religion and others turning to science.

(2018) document that in the 19th-century United States, a larger share of the population was organized in religious communities in counties that were exposed to higher common agricultural risk.

¹⁹Christian Science, founded in 1879, is part of the religious movements belonging to the metaphysical family. It seeks to restore the healing and thaumaturgic virtues of primitive Christianity and has been associated with avoidance of mainstream medicine (Stark, 1998).

²⁰On the other hand, a few groups of progressive Protestants called for a more scientific interpretation of the pandemic (Phillips, 2020).

3 Data

To conduct our analysis, we construct a novel dataset combining information on religiosity, scientific progress, and the incidence of the Great Influenza Pandemic. This section describes our outcome and main explanatory variables, with additional details provided in Appendix A. The first part of our analysis uses counties as the geographical unit of observation.²¹ The second part employs individual-level data. Table B.1 presents descriptive statistics for the main variables.

3.1 Measuring Religiosity

A key challenge in studying religiosity is the difficulty of measuring it accurately. It is particularly challenging to identify a metric that combines geographical granularity with high-frequency time variation.²²

In our analysis, we propose a novel measure of revealed religiosity based on newborn naming patterns. The underlying assumption is that parents who give comparatively more religious names are more likely to be religious themselves. Naming patterns thus provide a measure of parents’ “revealed religiosity” rather than that of the children themselves.²³

We now describe how we compute the religiosity score associated with first names. The key advantage of this approach is that it yields a geographically disaggregated yearly measure of religiosity and enables us to study its short- to medium-term dynamics. Our metric is conceptually similar to those developed by Abramitzky et al. (2016) and Andersen and Bentzen (2022), who measure religiosity based on whether children receive biblical, saint, or major religious-figure names. Our approach differs from theirs in that we *empirically* identify religious names using data on the entire population of newborns and existing indicators of religiosity.

3.1.1 Estimating Religiosity Scores for First Names

We use two main sources to compute religiosity scores. First, we construct county-cohort naming patterns from the full-count U.S. censuses between 1900 and 1930 (Ruggles, Flood, Foster, Goeken, Pacas, Schouweiler, and Sobek, 2021). Specifically, we extract the first names of all children born in the United States between 1896 and 1930 and aggregate them at the name-county-cohort level, obtaining a panel of name-county pairs at yearly frequency.²⁴ Second, we use county-level data from

²¹To address concerns regarding boundary changes over time, we use 1920 county definitions as our reference geography.

²²This data limitation is evident in historical settings—Squicciarini (2020), for instance, employs multiple measures of religiosity available for only a limited set of time points—but it also poses substantial constraints for contemporary studies.

²³A corollary is that names carry information about an individual’s religious background: while we cannot infer that an individual named “Paul” is comparatively more religious than one named “Harold,” we argue that Paul’s parents are likely more religious than Harold’s.

²⁴A cohort is defined as all babies born in a given year. The first cohort in our sample comprises all babies born in 1896. This is because the first Census of Religious Bodies was published in 1906, and we consider the ten cohorts preceding that year.

the Census of Religious Bodies. This census—conducted decennially between 1906 and 1936—allows us to construct, for each county and census decade, the share of people affiliated with any religious denomination.²⁵

To obtain the religiosity scores, we first compute the frequency of names. We denote with Name Frequency $_{cd}^k$ the number of children with name k born over the years $[d - 10, d)$ in county c . We then estimate the following regression:

$$y_{cd} = \alpha_c + \alpha_d + \sum_{k=1}^K \gamma^k \times \log(1 + \text{Name Frequency}_{cd}^k) + \varepsilon_{cd}, \quad (1)$$

where y denotes the share of people affiliated with any denomination in county c in decade d ; d corresponds to the two pre-pandemic decades of the religious censuses (1906 and 1916); and α_c and α_d are county and decade fixed effects, respectively.²⁶ The term K denotes the total number of names that occur in at least 0.3% of the overall sample.²⁷ To measure name frequencies, we include all babies born within ten years before each pre-pandemic religious census, thus restricting the sample to cohorts between 1896 and 1916. We then aggregate these frequencies from cohorts to decades to estimate equation (1).

We label the coefficient (γ^k) as the *religiosity score* associated with name k and interpret names with larger estimated religiosity scores ($\hat{\gamma}^k$) as conveying higher religiosity. Because equation (1) includes county fixed effects, larger religiosity scores are assigned to names that become comparatively more frequent in counties that experienced larger increases in religiosity. Figure 1 reports the estimated religiosity scores from equation (1), where the outcome variable is the share of people affiliated with any religious organization. The figure shows that religious-sounding names, such as “Joseph,” “Paul,” and “Elizabeth,” all exhibit positive and large estimated religiosity scores. Because our estimation method seeks to isolate *distinctively* religious names, relatively common names such as “Mary” or “John” are associated with negative scores. In the case of “Mary,” for instance, its popularity during

²⁵To gather information on the number of religious members in each county, reports were obtained directly from local churches and congregations. The shares are computed as the number of individuals affiliated with these groups divided by each county’s population. Throughout this period, Catholics and Protestants jointly account for more than 90% of those enumerated by the census. In the final Census of Religious Bodies (1936), some religious groups display lower reporting rates. Following Stark (1992), we consider the 1936 census less reliable and exclude it from our analysis.

²⁶In a robustness check, we compute an alternative measure of religiosity without fixed effects. The results are qualitatively the same. Moreover, the results are robust to using the raw name frequencies in equation (1) or, alternatively, applying an inverse hyperbolic sine transformation.

²⁷Following Fouka (2020), we restrict the number of names included in equation (1) primarily to avoid overfitting. Fouka (2020) uses a threshold of 1,000 occurrences for a name to be included in the analysis. In our preferred specification, we consider all names with a share in the overall sample of at least 0.3% and conduct robustness checks around this threshold. We include name frequencies in logs to reduce their skewness and add one to avoid dropping all counties where there is at least one name with no newborn children (approximately 40% of the sample).

this period implies that both religious and non-religious people used it for their kids, preventing it from being distinctively associated with religious individuals. Names with little connection to saints or biblical episodes are associated with negative religiosity scores. This is the case for Germanic names, such as “Bertha” and “Harold,” and other non-religious names, such as “Pearl.”

3.1.2 A Yearly County-Level Measure of Religiosity

From equation (1), we obtain a set of estimated religiosity scores $\{\hat{\gamma}^k\}_{k=1}^K$, which we use to construct a *yearly* indicator of religiosity at the county level.²⁸ Specifically, our measure of religiosity is defined as follows:

$$\hat{y}_{ct} = \sum_{k=1}^K \hat{\gamma}^k \times \log(1 + \text{Name Frequency}_{ct}^k), \quad (2)$$

where t captures cohorts between 1900 and 1930. For robustness checks, we also construct a series for Catholics and Protestants by considering religiosity scores associated with different denominations separately.

A potential concern regarding our religiosity indicator is how much variation in county religiosity names explain, net of that captured by fixed effects. In Appendix B, we provide two validation exercises for our measure. First, Figure B.1 compares our religiosity measure and the one obtained directly from the Census of Religious Bodies with two county-level scatterplots. Panel A shows in-sample correlations, comparing Census-measured and predicted religiosity in 1906 and 1916. Panel B compares the two measures in 1926. We refer to this as an “out-of-sample” correlation, as data from the Censuses of Religious Bodies conducted after the pandemic are not used to estimate religiosity scores. Both panels show a positive correlation between actual and predicted religiosity. This exercise provides evidence that naming patterns capture meaningful variation in religiosity.

Then, as alternative measures of religiosity, we use biblical and saint names from Abramitzky et al. (2016), as well as the county-level share of population with a religious affiliation (for all affiliations, and separately for Catholics and Protestants) recorded by the Census of Religious Bodies for 1906, 1916, and 1926.

3.2 Measuring Scientific Progress

We measure scientific progress at the local level using the share of individuals employed in STEM occupations. The rationale for this measure is that STEM occupations require science-oriented education, which plausibly correlates with more favorable attitudes toward science and scientific progress at the local level (Bianchi and Giorcelli, 2020; Biasi, Deming, Moser, and Dillon, 2022).

For each county and census year (1900 to 1930), we compute the share of individuals employed in

²⁸Note that by using the Census of Religious Bodies, we would only be able to get religiosity estimates at a decadal frequency. We validate our measure using the Census, below.

medical occupations and the share employed in STEM occupations. We compute these shares relative to (i) the entire population; (ii) the number of individuals employed in skilled occupations.²⁹

We complement our measure of scientific progress by using patent data from the Comprehensive Universe of U.S. Patents (CUSP; Berkes, 2018). The CUSP contains information on the universe of U.S. patents issued between 1836 and 2015. The data for the time period considered in our paper (1900–1930) are extracted from digitized patent documents obtained from the U.S. Patent and Trademark Office.

For the purpose of our analysis, we first assign each patent to a county based on an inventor’s residence and to a year based on the filing date. When a patent lists multiple inventors, we give equal weight to each inventor’s location. From the CUSP, we also collect the technology classes associated with each grant (according to the U.S. Patent Classification system) and assign them to technology groupings following the crosswalk provided by the National Bureau of Economic Research (Hall, Jaffe, and Trajtenberg, 2001).³⁰

3.3 Exposure to the Great Influenza Pandemic

To measure the incidence of the Great Influenza Pandemic across U.S. counties, we use mortality statistics assembled by the U.S. Department of Commerce. These were first collected in 1915 and, throughout the 1915–1918 period, they covered 40% of counties, accounting for more than 60% of the U.S. population. Following the methodology developed by Beach, Clay, and Saavedra (2022), we measure mortality caused by the influenza pandemic as average yearly deaths during the pandemic (1918–1919) divided by average yearly deaths in the years before the pandemic (1915–1917). Formally, excess mortality in county c is defined as

$$\text{Excess Deaths}_c = \frac{\frac{1}{2} \sum_{t=1918}^{1919} \text{Deaths}_{ct}}{\frac{1}{3} \sum_{t'=1915}^{1917} \text{Deaths}_{ct'}}. \quad (3)$$

²⁹The second measure increases the comparability of the control group with STEM workers. Table B.2 lists the set of occupations that we label as medical (Panel A), STEM (Panel B), and skilled (Panel C). By construction, medical occupations are also STEM occupations, and both are also skilled. The lists of Medical and STEM occupations are manually constructed. The list of skilled occupations includes all broad occupational groups that include at least one STEM occupation. These groups of occupations are: “Professional, Technical;” “Managers, Officials, and Proprietors;” and “Craftsmen.” Compared to other occupations, skilled occupations are characterized by a higher income score. Their occupational income score—a variable that assigns to each occupation the median total income of all persons with that occupation in 1950—is, on average, two-thirds greater than that of other occupations.

³⁰We combine the “chemical” and “drugs” NBER classes into a single class, which we label “pharmaceuticals.” We adopt this approach because of the large overlap between the chemical and drug patents at the county level, which would make it difficult to study them separately. All results for the pharmaceutical class also hold if we consider drug and chemical patents separately. An example of a pharmaceutical patent is shown in Figure B.2.

This measure represents our baseline treatment. We also report results using a binary treatment variable equal to one if the baseline treatment (Excess Deaths_c) is above its median and zero otherwise. Figure 2 displays the geographical variation in pandemic intensity in terms of excess deaths. The severity of the pandemic varies substantially across counties, even among geographically close ones. This measure relies on the assumption that higher mortality during the pandemic relative to the period just before is likely due to the pandemic itself.³¹

4 Main Results: County-Level Analysis

In this section, we present the baseline empirical strategy and show that exposure to the influenza pandemic led to an increase in both religiosity and scientific progress across counties.

4.1 Empirical Strategy

In the first part of the analysis, we separately study the impact of the pandemic on religiosity and scientific progress at the county level. Our sample consists of a panel of U.S. counties observed over the 1900–1929 period at decadal or yearly frequency. In particular, we leverage quasi-random variation in exposure to the pandemic across U.S. counties in a difference-in-differences (DiD) setting and estimate the following specification:

$$y_{ct} = \alpha_c + \alpha_{s(c) \times t} + \delta \times (\text{Post}_t \times \text{Excess Deaths}_c) + \beta \times \text{Post}_t \times \text{Population}_{c, 1910} + \varepsilon_{ct}, \quad (4)$$

where subscripts c and t denote county and time (decade or year), respectively; y_{ct} captures religiosity or scientific attitudes; α_c and $\alpha_{s(c) \times t}$ are county and state-by-time fixed effects; Post_t is an indicator variable equal to one if $t \geq 1918$ and zero otherwise; Excess Deaths_c measures the intensity of the pandemic in terms of excess deaths; and ε_{ct} is the error term. In all regressions, we control for an interaction term between population in 1910 and the post indicator ($\beta \times \text{Post}_t \times \text{Population}_{c, 1910}$). Standard errors are clustered at the county level. Our coefficient of interest, δ , captures the impact of the pandemic on religiosity or scientific sentiment. To investigate possible heterogeneity in treatment effects over time, we also estimate a more flexible specification that interacts Excess Deaths with

³¹A possible threat to this argument is the fact that WWI deaths might confound our results. However, in Figure B.3 we show that there is no significant correlation between WWI mortality and excess mortality. In Section 4, we show that our results are robust to controlling for a post-1918 time indicator interacted with WWI-related deaths.

biennial time dummies, rather than the Post indicator:³²

$$y_{ct} = \alpha_c + \alpha_{s(c) \times t} + \sum_{k \in \{1909, 1911, \dots, 1927\}} \delta^k \times I(t \in \{k, k+1\}) \times \text{Excess Deaths}_c + \beta \times \text{Post}_t \times \text{Population}_{c, 1910} + \varepsilon_{ct}, \quad (5)$$

where $I(t \in \{k, k+1\})$ is an indicator variable that takes value one if t is in the two-year window indexed by k , and zero otherwise.

A key identification concern is whether counties with differential exposure to the influenza pandemic would have experienced different outcome trends in its absence. While parallel trends after 1918 are ultimately untestable, we provide two complementary pieces of evidence in support of them. First, Table B.3 reports the correlation between pandemic intensity and religiosity, scientific attitudes, and a set of county covariates, all measured in 1910, the last census before the pandemic (or averaged over the decade 1901–1910 for yearly variables), accounting for population and state-level fixed effects.³³ Counties more exposed to the pandemic are observationally equivalent with respect to all variables except the share of foreigners. This aligns with the pandemic being comparatively more severe in urban areas, where immigrants were concentrated.

Second, we check whether control and treatment counties were on different trends before the shock by estimating event studies. Formally, in Equation (5), this implies that the estimates of δ^k should not be statistically different from zero before the pandemic hit, i.e., for all $k < 1917$.³⁴ Our estimates support the parallel-trends assumption.

However, even with parallel trends, our estimates could be confounded by shocks correlated with both pandemic intensity and our outcomes. Possible candidates are WWI deaths or local economic conditions, such as unemployment. To address these concerns, we conduct a battery of robustness checks for each analysis, controlling for potential confounders (e.g., unemployment) and testing alternative specifications. These checks are described in detail below.

³²In the dynamic DiD specifications, we code periods over two-year windows to reduce noisy fluctuations in the dependent variables and improve efficiency by pooling observations.

³³State fixed effects control for the fact that the pandemic spread from East to West between August 1918 and November 1918.

³⁴Since the setting is not staggered—because the pandemic hit each county in the same period—equations (4) and (5) can be estimated through standard two-way fixed effects (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). Callaway, Goodman-Bacon, and Sant’Anna (2021), however, caution against using continuous treatments. We code a binary indicator equal to one for counties with above-median excess deaths. Throughout the paper, we show that the continuous and binary treatments yield qualitatively similar results.

4.2 The Effect of the Influenza Pandemic on Religiosity

Table 1 shows the DiD estimates of the effect of the influenza pandemic on religiosity. In columns (1–3), we use data from the Census of Religious Bodies to measure the share of individuals affiliated with any religious denomination, Catholic denominations, or Protestant denominations, respectively. The estimates suggest that counties comparatively more exposed to the pandemic experienced an increase in religiosity, with no significant differences between Catholics and Protestants.

This measure of religiosity is available at the decade level (1906, 1916, and 1926) and therefore does not allow us to study high-frequency variation in religiosity. In columns (4–6), we turn to our name-based measure of religiosity. This variable allows us to measure county-level religiosity between 1900 and 1929 at a yearly frequency. The results suggest that moving from a county at the 25th percentile of the excess mortality distribution to one at the 75th percentile led to an increase in religiosity of 10%.

Figure 3 reports the coefficients from the event study. The estimates support the patterns observed in the DiD analysis and confirm the absence of pre-trends. The increase in religiosity appears to persist over the decade after the pandemic. These findings are consistent with the literature documenting substantial persistence in religiosity (e.g., Squicciarini, 2020).

Robustness Checks We perform several exercises to assess the robustness of these findings.³⁵ First, weighting regressions by 1910 county population confirms that our results are not driven by small counties (Table B.4). Second, the baseline findings remain robust to coding treatment as a binary variable and controlling for WWI mortality (Table B.5, columns 1–2).

A key potential concern is that internal migration could confound our results if comparatively more religious individuals moved into counties where the pandemic had been more severe, perhaps motivated by slacker labor markets. We address this concern in two ways. First, we exclude from the name-based religiosity measure all individuals who, in the 1930 census, reside in a state different from their birth state. The results remain unchanged (Table B.5, column 3). Second, we compute a county-decade measure of religiosity based solely on the names of the adult population. The in-migration mechanism would predict a positive impact of the pandemic on this variable. However, we find no evidence of any such effect (Table B.5, column 4).

We also verify that our findings are robust to alternative specifications of our religiosity measures (Table B.6). First, one concern is that firstborns are often named after deceased grandparents, potentially reflecting earlier generations' religiosity rather than parents' attitudes. If higher mortality in pandemic-affected areas increased the likelihood of recently lost grandparents, our results might reflect

³⁵ A summary of all robustness checks is provided in Appendix C.

a mechanical effect. Dropping firstborns leaves our findings unchanged (column 1). Similarly, results hold when dropping children beyond the fourth birth order (column 2) and when computing within-household average religiosity to account for differential fertility across religious households (column 3). Table B.7 shows that our measure is robust to estimating religiosity scores without county fixed effects (columns 1–2) and to varying the frequency threshold for name inclusion from our baseline of 0.3% (columns 3–8).

As an alternative name-based indicator of religiosity, we follow Abramitzky et al. (2016) and use biblical and saint names to proxy for religiosity. Table B.8 shows that the (log) number of children with biblical or saint names increased in areas more exposed to the pandemic after 1918. Figure B.4 demonstrates that this alternative measure is positively correlated with our baseline religiosity measure at the county level.

To ensure that our results are not driven by specific geographic areas, we perform two exercises. First, we exclude one state at a time from the sample; the estimates remain remarkably stable and significant throughout (Figure B.5a). Second, to verify that our findings extend beyond rural areas, we replicate our analysis at the city level using data on city-level deaths from Clay, Lewis, and Severnini (2019).³⁶ The name-based religiosity measure is computed by leveraging variation in the naming patterns of children born in each city. Column 1 of Table B.9 shows that religiosity increased in cities more severely affected by the pandemic, consistent with our baseline county-level analysis. Moreover, the city-level sample includes several cities in Southern states not included in our main analysis due to a lack of county-level mortality data.³⁷

Panel A of Figure B.7 shows the sensitivity of our main coefficient to the inclusion of various controls, including unemployment.³⁸ We add each control individually to the baseline specification, measuring it at its 1910 value (the last census before the pandemic) as a share of the 1910 population and interacting it with the post-influenza indicator. The coefficients remain virtually unchanged.

Finally, we assess the validity of the parallel trends assumption using the approach of Rambachan and Roth (2023). Figure B.8a compares the confidence intervals of the event-study estimates with those obtained when allowing for per-period violations from linear trends up to a specific threshold (M). The estimates remain significant at the 90% level for values of M below 0.1, suggesting that our results are robust to low power of pre-trends tests and nonlinear violations of parallel trends.

Overall, we document that the pandemic increased religiosity across different specifications, samples, and measures. This finding is consistent with the religious coping hypothesis, which posits that religion may serve as a coping mechanism for mental and psychological distress (e.g., Pargament,

³⁶This dataset is discussed in more detail in Appendix A.8.

³⁷Figure B.6 shows the number of cities included in the sample by state and their location.

³⁸We measure unemployment as the share of individuals outside the labor force.

2001; Bentzen, 2019, 2021). A complementary explanation is that individuals turn to religion as an insurance mechanism against the negative effects of the pandemic. In Section 6, we discuss both explanations in more detail.

4.3 The Effect of the Influenza Pandemic on Scientific Progress

In this section, we study how the influenza pandemic impacted attitudes towards science. We show that this shock led to an increase in the share of people in STEM occupations and in patenting activity.

We estimate equation (4) using the share of individuals employed in STEM relative to the overall population as the dependent variable (Table 2, column 1). We perform this analysis at the decade level since this measure is taken from population censuses (1900, 1910, 1920, and 1930). We document an increase in the share of workers in STEM occupations in counties more severely hit by the pandemic. Moving from the 25th to the 75th percentile of the excess mortality distribution leads to a 17% increase in the share of individuals in STEM occupations. In columns 2 and 3, we estimate the same specification, using the share of individuals in medical occupations and the share in STEM non-medical occupations as the dependent variable, respectively. The estimates suggest that our results are not exclusively driven by medical occupations. Individuals reacted to the pandemic by turning to occupations related to medicine, as well as to other scientific fields. Columns 4–6 study the effect of the pandemic on STEM occupations when only considering the skilled sub-sample of the population. The results remain similar, suggesting that the pandemic induced entry into scientific occupations specifically, rather than a general shift toward skilled employment.

In Table 3, we study whether this increase in scientific occupation also translated into more innovation by focusing on county-level patent production. These data allow us to construct a high-frequency indicator of scientific activity. Column 1 of Table 3 reports the results of estimating equation (4) on the total volume of innovation—measured as $\log(1 + \text{number of patents})$ in a given county-year. We find that moving from a county at the 25th percentile to one at the 75th percentile of the excess-deaths distribution leads, on average, to an increase of 6.5% in the number of patents granted by county year. In column 2, we estimate the same specification on $\log(1 + \text{number of patents in pharmaceuticals})$, and, in column 3, we estimate the same specification on $\log(1 + \text{number of patents not in pharmaceuticals})$. The impact of the shock is positive in both cases, but larger in pharmaceuticals than in other fields: the effect relative to the mean is nearly twice as large (11% compared to 6%). In columns 4–6, we estimate the same specifications of columns 1–3 on the subset of counties that produced at least one patent a year. The magnitude is larger, suggesting that the increase in patenting comes both from the extensive margin and from the intensive margin.

Finally, Figure 4 shows the results shown in column 1 of Table 3 in an event-study framework. The coefficients suggest that the number of patents granted after the pandemic increased significantly in

more exposed counties, implying that the pandemic induced a sizable increase in innovation activity that persisted for at least one decade after the shock.³⁹

Robustness Checks We perform several exercises to assess the robustness of these findings. First, we verify that small counties with relatively low scientific production do not drive our results. Tables B.10 and B.11 replicate the results of Tables 2 and 3, respectively, weighting each county by 1910 population. The results remain qualitatively the same. Table B.12 addresses the potentially confounding role of immigration by excluding from the sample all individuals who, in the 1930 census, reside in a state different from their birth state. The results remain unchanged.

We also verify the robustness of the patent-based findings (Table B.13). The baseline results hold when coding treatment as a binary variable and controlling for WWI deaths interacted with the post-treatment indicator (columns 1–2). In the baseline specification, we take the logarithm of the number of patents and add one to avoid dropping zeros. Columns 3–4 use alternative transformations of the dependent variable and obtain quantitatively similar findings. Additionally, estimating the baseline regression model with a Poisson Quasi-Maximum Likelihood estimator yields similar results (column 5).

A possible concern is that our results may be driven by low-quality innovation. Newspapers of the day often advocated remedies for influenza that were not science- or evidence-based, some of which may have been patented. To address this concern, we use the text-based measures of “breakthroughness” developed by Kelly, Papanikolaou, Seru, and Taddy (2021).⁴⁰ We assign to every patent a dummy equal to one if the patent’s importance is in the top 20% of the distribution and zero otherwise. Table B.14 reports the results. The number and share of these breakthrough patents substantially increase in counties hit harder by the pandemic.

To ensure that our results are not driven by specific geographic areas, we exclude one state at a time from the sample. The coefficient remains stable (Figure B.5b). While most innovation activity clusters in urban areas, our baseline analysis uses counties as the unit of observation. To verify that the results do not simply reflect rural-urban disparities, we estimate the effect of the pandemic on innovation at the city level. Column 2 in Table B.9 reports estimates of equation (4) for the panel of cities described in Section A.8. The results confirm the county-level evidence: despite the smaller sample size, we find a positive and statistically significant effect of the pandemic on innovation production.

³⁹The timing of the effect is in line with those described in Furman, Nagler, and Watzinger (2021) and Berkes and Nencka (Berkes and Nencka). These papers find that patent repositories and Carnegie libraries, respectively, begin to have a positive and statistically significant impact on local patenting activity within the first year.

⁴⁰As discussed by Berkes (2018) and Andrews (2021), citation-based quality measures during this period are noisy and mostly uninformative due to the lack of a mandatory reference section until 1947. The measure built by Kelly et al. (2021) identifies important patents based on the textual similarity of a given patent to previous and subsequent work. Important patents are those that are distinct from previous work but similar to subsequent innovations.

Panel B of Figure B.7 shows the sensitivity of our main coefficient to the inclusion of various controls, including unemployment.⁴¹ We add each control individually to the baseline specification, measuring it at its 1910 value (the last census before the pandemic) as a share of the 1910 population and interacting it with the post-influenza indicator. The coefficients remain virtually unchanged.

Finally, Figure B.8b confirms that the results remain remarkably robust to low-power pre-trends tests and nonlinear violations of parallel trends.

Overall, the results presented in this section suggest that the pandemic increased scientific attitudes across different specifications, samples, and indicators. The shock increased the propensity to select medical and, more broadly, STEM occupations. On the innovation side, the pandemic spurred growth in both overall and pharmaceutical innovation, possibly in an attempt to mitigate the negative effects of the pandemic or to work on preventing future health crises.

4.4 Joint Dynamics of Religiosity and Innovation

Having studied the impact of the pandemic on religiosity and scientific attitudes separately, we now turn our attention to their joint evolution. Specifically, we test whether the same counties experienced increases in both religiosity and scientific attitudes, or whether these effects are driven by different sets of counties.

Columns 1 and 2 of Table B.15 report the correlation between innovation and religiosity before and after the pandemic, respectively, controlling for county and state-by-year fixed effects, following Bénabou et al. (2022). The relationship shifts from negative before the pandemic to positive after the pandemic, as shown in Figure 5 and confirmed in column 3, which pools observations across the pre- and post-pandemic periods.

We then estimate the following regression model:

$$\begin{aligned}
y_{ct} = & \alpha_c + \alpha_{s(c) \times t} + \delta_1 \times (\text{Excess Deaths}_c \times \text{Post}_t) + \delta_2 \times \text{Religiosity}_{ct} + \\
& + \delta_3 \times (\text{Religiosity}_{ct} \times \text{Post}_t) + \delta_4 \times (\text{Religiosity}_{ct} \times \text{Excess Deaths}_c) + \\
& + \delta_5 \times (\text{Excess Deaths}_c \times \text{Post}_t \times \text{Religiosity}_{ct}) + \beta \times \text{Post}_t \times \text{Population}_{c,1910} + \varepsilon_{ct},
\end{aligned} \tag{6}$$

where y_{ct} is the number of patents divided by county population in 1910, as in Bénabou et al. (2015), and Religiosity_{ct} is the religiosity measure described in Section 3.1. The coefficient δ_1 measures the impact of the pandemic on scientific progress, δ_2 captures the relationship between patenting and religiosity in the pre-pandemic period for counties with zero excess deaths, δ_4 captures how this relationship differs with county exposure to the pandemic, and δ_5 captures how this correlation changed after 1918 as a function of exposure to the pandemic. As in previous specifications, we also control

⁴¹We measure unemployment as the share of individuals outside the labor force.

for the interaction between county population in 1910 and a post-treatment indicator.

In column 4, we estimate regression (6) and find that this shift in the correlation between religiosity and innovation co-moved with county-level exposure to the pandemic. These results suggest that counties comparatively more affected by the pandemic experienced a joint increase in religiosity and innovation. In Section 5, we use individual-level data to study the mechanisms underlying these results.

5 Mechanisms: Individual-Level Analysis

We now investigate how individual responses within counties can account for the contemporaneous increase in religiosity and scientific activity documented above. We examine whether the same individuals within counties drove the increases in both religiosity and scientific orientation, or whether different individuals responded differently. We also identify which segments of the population turned to religion versus science. To do so, we leverage individual-level data and focus on individuals who were heads of household in the 1930 census.⁴² More precisely, we proceed in three steps: first, we examine whether the pandemic widened differences in religiosity between science-oriented individuals and the rest of the population; second, we study how individuals' religious backgrounds shaped their occupational and religious responses to the shock; and third, we assess whether these heterogeneous responses translated into increased religious polarization within counties.

We document three main findings. First, the pandemic widened the distance in religiosity between individuals in STEM occupations and the rest of the population. Second, individuals from less religious backgrounds became more likely to select a STEM occupation, while those from more religious backgrounds increased their religiosity. Third, we observe that the pandemic increased religious polarization.

Taken together, these three results suggest that the pandemic shock led to different reactions within society—based, for instance, on individuals' initial scientific orientation or religious background—with people becoming even more distant in their religiosity than they were before the pandemic. This within-county analysis reveals important heterogeneity in how individuals react to a negative shock and helps reconcile our aggregate findings with the existing literature on the negative relationship between religion and scientific progress.

⁴²The “head of household” variable is provided by the census. During this period, the father and/or husband was usually considered the head of the household whenever present.

5.1 The Pandemic Increased the Distance in Religiosity Between Science-Oriented Individuals and the Rest of the Population

We first examine the impact of the influenza pandemic on the religiosity of science-oriented individuals. Using full-count census data, we compute our measure of revealed religiosity for each individual based on the names of their children. Formally, we estimate the following triple-differences model:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(i) \times \text{STEM}(i)} + \alpha_{t \times \text{STEM}(i)} + \alpha_{c(i) \times t} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \delta \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_t \times \text{STEM}_i) + \varepsilon_{it}, \end{aligned} \quad (7)$$

The sample includes all children born in the U.S. between 1900 and 1929.⁴³ Each observation (a child) is weighted by the inverse of the number of children in the household.⁴⁴ The dependent variable is the religiosity associated with a child's name born in household i , living in county $c(i)$, in birth year t . All specifications include county-by-occupation ($\alpha_{c(i) \times \text{STEM}(i)}$), year-by-STEM occupation ($\alpha_{t \times \text{STEM}(i)}$), and county-by-year ($\alpha_{c(i) \times t}$) fixed effects. $\mathbf{x}_i$ is a vector of urban status, race, and fixed effects for the birthplace of the household head. The coefficient δ compares STEM and non-STEM households, before and after the pandemic, by county-level exposure to the pandemic. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_t is an indicator variable equal to one if the child was born in or after 1918, and zero otherwise. STEM_i is an indicator variable taking value one if the household head is employed in a STEM occupation in the 1930 census, and zero otherwise.

Table 4 shows the results. In column 1, we report results for the full population. In column 2, we restrict attention to heads of households employed in skilled occupations. In the aftermath of the pandemic, individuals in STEM became less religious than non-STEM individuals in counties more exposed to the shock. The effect is remarkably stable within the sample of skilled occupations, suggesting that the widening gap in religiosity does not simply reflect broader socio-economic disparities within society.

These results reveal that, while at the county level we observe a contemporaneous increase in religiosity and scientific attitudes, the pattern differs when we examine individuals' reactions *within* counties. Specifically, in response to the pandemic, the gap in religiosity between science-oriented individuals and the rest of the population increased, even when comparing STEM workers to other skilled occupations. This suggests that, within the counties most affected by the influenza pandemic, *different* individuals drove increases in religiosity and scientific attitudes.

⁴³We restrict the analysis to U.S.-born individuals as we only assign background religiosity to individuals born in the United States.

⁴⁴In other words, each household in our regression has a weight of one.

5.2 Turning to Science or Turning to Religion

In this section, we document that differences in individuals’ preexisting religious backgrounds shaped their response to the influenza pandemic. From full-count census data, we construct a measure of “background religiosity” for each individual based on their own name. Specifically, we interpret an individual’s own name as conveying information about their parents’ religiosity and, thus, the individual’s religious background.

We first show that an individual’s religious background shaped their propensity to work in a scientific occupation following the pandemic. To measure this, we use an indicator equal to one if an individual was employed in a STEM occupation and zero otherwise. Next, we document that an individual’s religious background also shaped their response to the pandemic in terms of religiosity. We estimate two triple-difference specifications, one for the likelihood of selecting a STEM occupation and one for revealed religiosity.

To investigate the heterogeneous impact of the pandemic on occupational choice, we estimate the following equation:

$$\begin{aligned} \text{STEM}_i = & \alpha_{c(i) \times t(i)} + \alpha_{c(i) \times B(i)} + \alpha_{B(i) \times t(i)} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \delta_1 \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_i \times \text{High Religious Background}_i) + \varepsilon_i, \end{aligned} \quad (8)$$

where i denotes an individual i born in year $t(i)$. The unit of observation is an adult observed once in the 1930 census. The dependent variable is an indicator equal to one if the head of household i is employed in a STEM occupation in 1930 and zero otherwise. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_i is an indicator equal to one for individuals aged between 18 and 25 in 1918 and zero otherwise. We consider individuals as treated if they were 18 to 25 years old at the beginning of the pandemic for two reasons. First, according to the “impressionable years” hypothesis—a long-standing argument in psychology—economic, social, and cultural attitudes and beliefs are formed during early adulthood, approximately between the ages of 18 and 25, and change slowly after then (Giuliano and Spilimbergo, 2023). Second, the young cohort was plausibly more likely to adjust their employment choices as they were about to enter (or had recently entered) the labor market. The background religiosity term $\text{High Religious Background}_i$ is an indicator variable taking value one if the religiosity score of individual i ’s name is in the top 50% of the overall distribution and zero otherwise.

Similarly, to study the impact of the pandemic on religiosity, we estimate the following equation:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(i) \times t} + \alpha_{c(i) \times B(i)} + \alpha_{B(i) \times t} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \delta_2 \times (\text{Excess Deaths}_{c(i)} \times \text{Post}_{it} \times \text{High Religious Background}_i) + \varepsilon_{it}, \end{aligned} \quad (9)$$

The sample includes children born between 1900 and 1929, and each observation (a child) is weighted by the inverse of the number of children in each household. The dependent variable Religiosity_{it} denotes the religiosity score of a child born in year t in household i , living in county $c(i)$. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_{it} is a categorical variable equal to one for children born after 1918 and zero otherwise. $\text{High Religious Background}_i$ is an indicator variable taking value one if the average religiosity of the head of household is equal to or above the median of background religiosity, or zero otherwise.

In equations (8) and (9), the terms $\alpha_{c(i) \times t}$, $\alpha_{c(i) \times B(i)}$, and $\alpha_{B(i) \times t}$ denote county-by-year, county-by-high religious background, and year-by-high religious background fixed effects, respectively, and $\mathbf{x}_i$ includes urban status, race, and fixed effects for the birthplace of the household head. The coefficients δ_1 and δ_2 quantify the effect of county-level exposure to the pandemic, comparing individuals in the top 50% of the background religiosity distribution with the rest of the population on STEM employment and religiosity, respectively.

Table 5 presents the results of the analysis. In columns 1 and 3, the outcome variable is a dummy equal to one if the head of household is employed in a STEM occupation and zero otherwise. In columns 2 and 4, the dependent variable is the individual's own religiosity. The main coefficients of interest are δ_1 and δ_2 . The findings suggest that an individual's religious background influenced their response to the pandemic. Specifically, those raised in less religious households became more likely to choose a scientific occupation in the face of adversity (column 1), whereas those raised by more religious parents became more likely to turn to religion (column 2). Notably, these patterns persist even when restricting the sample to individuals in skilled occupations (columns 3 and 4). This suggests that individuals' religious backgrounds shaped their response to the pandemic not only across socio-economic groups, but also within groups characterized by relatively high levels of human capital and economic resources. Additionally, we find no significant association between religiosity and the choice of skilled occupations overall (Table B.16, column 2).

The increase in STEM occupations following the pandemic may have been partly driven by increased demand for scientific occupations, particularly in medicine and public health. However, demand-side factors alone do not explain the systematic sorting by religious background that we document. Even when controlling for socio-economic observable characteristics and restricting our sample to skilled occupations, those from less religious backgrounds were significantly more likely to select into STEM fields. This pattern suggests that supply-side factors, specifically individuals' preexisting cultural and religious orientations, played a role in shaping occupational responses to the pandemic.

5.3 The Polarization of Religiosity

The analyses we have presented so far in this section document that individuals from different religious backgrounds responded differently to the pandemic: Those from less religious backgrounds turned to science, while those from more religious backgrounds increased their religiosity. This divergent response suggests the possibility of increased polarization within counties. To investigate this, we examine how the pandemic affected the entire distribution of religiosity, focusing on whether individuals at different points of the religious background distribution experienced differential changes in religiosity.

We divide the distribution of background religiosity into five quintiles, which we label Q , and estimate the following equation:

$$\begin{aligned} \text{Religiosity}_{it} = & \alpha_{c(i) \times t} + \alpha_{c(i) \times Q(i)} + \alpha_{Q(i) \times t} + \mathbf{x}'_i \boldsymbol{\beta} + \\ & + \sum_{\substack{k=1 \\ k \neq 3}}^5 \delta_k \times [\text{Excess Deaths}_{c(i)} \times \text{Post}_{it} \times I(Q_i = k)] + \varepsilon_{it}, \end{aligned} \quad (10)$$

The sample includes all children born between 1900 and 1929. Each observation (a child) is weighted by the inverse of the number of children in each household. The dependent variable measures the religiosity score associated with the name of a child born in household i living in county $c(i)$ in year t . The equation includes county-by-year ($\alpha_{c(i) \times t}$), county-by-quintiles of religious background ($\alpha_{c(i) \times Q(i)}$), and quintiles of religious background-by-year ($\alpha_{Q(i) \times t}$) fixed effects. The term $\mathbf{x}_i$ is a vector of urban status, race, and fixed effects for the birthplace of the household head. $\text{Excess Deaths}_{c(i)}$ measures the intensity of the pandemic in terms of excess deaths, as explained in Section 3.3. Post_{it} is an indicator equal to one if the child was born in or after 1918, and zero otherwise. For each quintile k , $I(Q_i = k)$ is a dummy variable taking value one if the household background religiosity is in the k -th quintile of the distribution of background religiosity, equal to zero otherwise.

In Figure 6, we report the estimated $\{\delta_k\}$ coefficients.⁴⁵ For each quintile of background religiosity k , the coefficient δ_k quantifies the impact of county-level exposure to the pandemic on the change in religiosity of individuals in quintile k relative to those in the third quintile (our baseline). The results provide clear evidence of increased polarization. Individuals from higher quintiles of background religiosity (i.e., those raised in more religious households) experienced larger increases in religiosity in response to the pandemic. The effect is monotonic across quintiles: as background religiosity moves from the lowest to the highest quintile, the pandemic's effect on religiosity becomes progressively larger. This pattern indicates that, within the same county, individuals from different religious back-

⁴⁵The associated regression estimates are reported in Table B.17.

grounds became more divergent in their religiosity, increasing the polarization of religious sentiment within society.⁴⁶

The individual-level results we presented in this section help us uncover the channels behind the contemporaneous increase in religiosity and scientific attitudes at the county level. They suggest that within counties, individuals reacted differently to the same shock based on their pre-pandemic scientific orientation or religious background. As a result, while a county may have become more religious *and* more scientifically oriented, individuals seemed to turn either to religion *or* to science, leading to within-county polarization of religiosity.

6 Discussion: Interpretation and Limitations of the Results

Our analysis highlights two main findings. First, the 1918–1919 influenza pandemic led to an increase in both religiosity and scientific attitudes across U.S. counties, such that counties hit harder by the shock became simultaneously more religious and more scientifically oriented. Second, this aggregate pattern masks substantial heterogeneity at the individual level: within counties, some individuals responded to the shock by turning toward religion, while others turned toward science.

A natural concern is that other factors related to the pandemic may confound these patterns. We address this in several ways. We show that initial religiosity, scientific progress, and other demographic characteristics are not systematically related to pandemic intensity. Event-study analyses indicate the absence of pre-trends, suggesting that treated and control counties were on similar trajectories prior to 1918. In addition, our results are robust to accounting for potentially confounding factors such as differential fertility, World War I mortality, and migration patterns. Taken together, the empirical evidence, supported by historical accounts, makes it difficult to attribute the observed changes in religiosity and scientific activity to factors other than the pandemic itself.

A second concern relates to the measurement of religiosity and scientific progress. We show that our results are robust to alternative ways of constructing the name-based measure and to alternative classifications of religious names. Moreover, in counties hit harder by influenza, the share of people affiliated with a religious denomination increased, providing independent evidence that the pandemic led to higher local religiosity. Similarly, we measure scientific attitudes using both the share of individuals employed in scientific occupations and patenting activity. Despite their limitations, both measures consistently point to an increase in scientific engagement following the pandemic.

After documenting these aggregate patterns, a natural question is how they can be reconciled with the negative relationship between religion and science documented in the literature. Our individual-level evidence indicates that the aggregate patterns are not driven by the same individuals becoming

⁴⁶The correlation between revealed religiosity and background religiosity is equal to 0.13 and highly statistically significant ($p < .001$), in line with a large literature on cultural transmission (Bisin and Verdier, 2001).

simultaneously more religious and more science-oriented. Instead, different individuals responded to the same adverse shock in different ways. The pandemic widened the gap in religiosity between science-oriented individuals and the rest of the population. Moreover, individuals from less religious backgrounds became more likely to select into scientific occupations, while individuals from more religious backgrounds became more religious. These findings imply that religion and science functioned as alternative responses to adversity across different segments of the population.

Several mechanisms are consistent with the heterogeneous responses we document. Individuals may turn to religion as a coping device following a negative shock, in line with the religious coping hypothesis (Bentzen, 2019), or as a form of social insurance. The persistence of increased religiosity up to ten years after the pandemic suggests that these responses reflect lasting changes rather than transitory reactions. On the scientific side, individuals may have sought to mitigate the effects of the pandemic, respond to psychological distress, or take advantage of increased labor demand in scientific and medical occupations.

While our analysis is not designed to disentangle these mechanisms, the results point to a clear and robust pattern: individuals' preexisting religious backgrounds played a central role in shaping their responses to the pandemic. These patterns persist even within the sample of skilled occupations, suggesting that religious background is not simply proxying for human capital or socioeconomic status. Rather than isolating a single mechanism, our contribution is to document how a common adverse shock generated heterogeneous individual responses and increased polarization within communities, as different individuals turned to either religion or science.

7 Conclusions

This paper studies how individuals and communities responded to the 1918–1919 influenza pandemic in the United States. We show that counties hit harder by the pandemic experienced contemporaneous increases in religiosity and in science-oriented activity (as reflected in occupational choices). This pattern seems to contradict evidence of a negative relationship between religiosity and science.

We reconcile this tension by documenting heterogeneity in individual responses. The joint increase at the county level does not reflect the same individuals becoming both more religious and more science-oriented. Instead, individuals from more religious backgrounds intensified their religiosity, while individuals from less religious backgrounds were more likely to turn toward science. As a result, the pandemic widened pre-existing differences and increased within-county divergence.

These findings highlight the limits of interpreting aggregate responses to large adverse shocks. Average changes in religiosity or scientific activity can mask substantial heterogeneity in behavior—and, in this setting, that heterogeneity translated into greater social division despite increases in both religiosity and scientific activity.

While our analysis focuses on a historical shock, the mechanism may also help interpret responses to other large, high-salience adverse events. Our results suggest that such shocks can deepen social divisions, implying that polarization may arise not only from slow-moving ideological change but also from acute shocks that push different groups in different directions.

References

- Abramitzky, R., L. P. Boustan, and K. Eriksson (2016). Cultural assimilation during the age of mass migration. *NBER Working Paper* (No. w22381).
- Agarwal, R. and P. Gaule (2022). What Drives Innovation? Lessons from COVID-19 R&D. *Journal of Health Economics* 82, 102591.
- Ager, P. and A. Ciccone (2018). Agricultural Risk and the Spread of Religious Communities. *Journal of the European Economic Association* 16(4), 1021–1068.
- Ager, P., C. W. Hansen, and L. Lønstrup (2016). Church Membership and Social Insurance: Evidence from the Great Mississippi Flood of 1927. *Southern Denmark University Discussion Papers on Business and Economics*.
- Almond, D. (2006). Is the 1918 Influenza Pandemic Over? Long-term Effects of in Utero Influenza Exposure in the Post-1940 US Population. *Journal of Political Economy* 114(4), 672–712.
- Andersen, L. H. and J. S. Bentzen (2022). In the Name of God! Religiosity and the Transition to Modern Economic Growth. *CEPR Discussion Paper* (DP16938).
- Andrews, M. (2021). Historical Patent Data: A Practitioner’s Guide. *Journal of Economics and Management Strategy* 30(2), 368–397.
- Ano, G. G. and E. B. Vasconcelles (2005). Religious Coping and Psychological Adjustment to Stress: A Meta-Analysis. *Journal of Clinical Psychology* 61(4), 461–480.
- Autor, D., D. Dorn, G. Hanson, and K. Majlesi (2020). Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. *American Economic Review* 110(10), 3139–3183.
- Babina, T., A. Bernstein, and F. Mezzanotti (2023). Financial Disruptions and the Organization of Innovation: Evidence from the Great Depression. *The Review of Financial Studies* 36(11), 4271–4317.
- Barro, R. J. (2022). Non-pharmaceutical Interventions and Mortality in US Cities During the Great Influenza Pandemic, 1918–1919. *Research in Economics* 76(2), 93–106.
- Barro, R. J., J. F. Ursúa, and J. Weng (2020). The Coronavirus and the Great Influenza Pandemic: Lessons from the “Spanish Flu” for the Coronavirus’s Potential Effects on Mortality and Economic Activity. *NBER Working Paper* (No. w26866).
- Barry, J. M. (2020). *The Great Influenza: The Story of the Deadliest Pandemic in History*. London (UK): Penguin.

- Bazzi, S., M. Fiszbein, and M. Gebresilas (2020). Frontier Culture: The Roots and Persistence of “Rugged Individualism” in the United States. *Econometrica* 88(6), 2329–2368.
- Beach, B., K. Clay, and M. Saavedra (2022). The 1918 Influenza Pandemic and Its Lessons for COVID-19. *Journal of Economic Literature* 60(1), 41–84.
- Becker, S. O., J. Rubin, and L. Woessmann (2021). Religion in Economic History: A Survey. *The Handbook of Historical Economics*, 585–639.
- Becker, S. O. and L. Woessmann (2009). Was Weber Wrong? A Human Capital Theory of Protestant Economic History. *The Quarterly Journal of Economics* 124(2), 531–596.
- Belloc, M., F. Drago, and R. Galbiati (2016). Earthquakes, Religion, and Transition to Self-Government in Italian Cities. *The Quarterly Journal of Economics* 131(4), 1875–1926.
- Bénabou, R., D. Ticchi, and A. Vindigni (2015). Religion and Innovation. *American Economic Review: Papers & Proceedings* 105(5), 346–351.
- Bénabou, R., D. Ticchi, and A. Vindigni (2022). Forbidden Fruits: The Political Economy of Science, Religion and Growth. *The Review of Economic Studies* 89(4), 1785–1832.
- Bentzen, J. (2019). Acts of God? Religiosity and Natural Disasters Across Subnational World Districts. *The Economic Journal* 126(622), 2295–2321.
- Bentzen, J. S. (2021). In Crisis, We Pray: Religiosity and the COVID-19 Pandemic. *Journal of Economic Behavior & Organization* 192, 541–583.
- Berkes, E. (2018). Comprehensive Universe of US Patents (CUSP): Data and Facts. *Working Paper*.
- Berkes, E., O. Deschenes, R. Gaetani, J. Lin, and C. Severen (2023). Lockdowns and Innovation: Evidence from the 1918 Flu Pandemic. *Review of Economics and Statistics*, 1–30.
- Berkes, E. and P. Nencka. Knowledge access: The effects of carnegie libraries on innovation. *The Review of Economics and Statistics*.
- Bianchi, N. and M. Giorelli (2020). Scientific Education and Innovation: from Technical Diplomas to University STEM Degrees. *Journal of the European Economic Association* 18(5), 2608–2646.
- Biasi, B., D. Deming, P. Moser, and E. W. Dillon (2022). *Education and Innovation*, pp. 537–558. Chicago: University of Chicago Press.
- Biavaschi, C., C. Giuliatti, and Z. Siddique (2017). The Economic Payoff of Name Americanization. *Journal of Labor Economics* 35(4), 1089–1116.
- Bisin, A. and T. Verdier (2001). The Economics of Cultural Transmission and the Dynamics of Preferences. *Journal of Economic Theory* 97(2), 298–319.
- Botticini, M. and Z. Eckstein (2012). *The Chosen Few: How Education Shaped Jewish History, 70–1492*. Princeton University Press.
- Boustan, L. P., M. E. Kahn, and P. W. Rhode (2012). Moving to Higher Ground: Migration Response to Natural Disasters in the Early Twentieth Century. *American Economic Review* 102(3), 238–244.

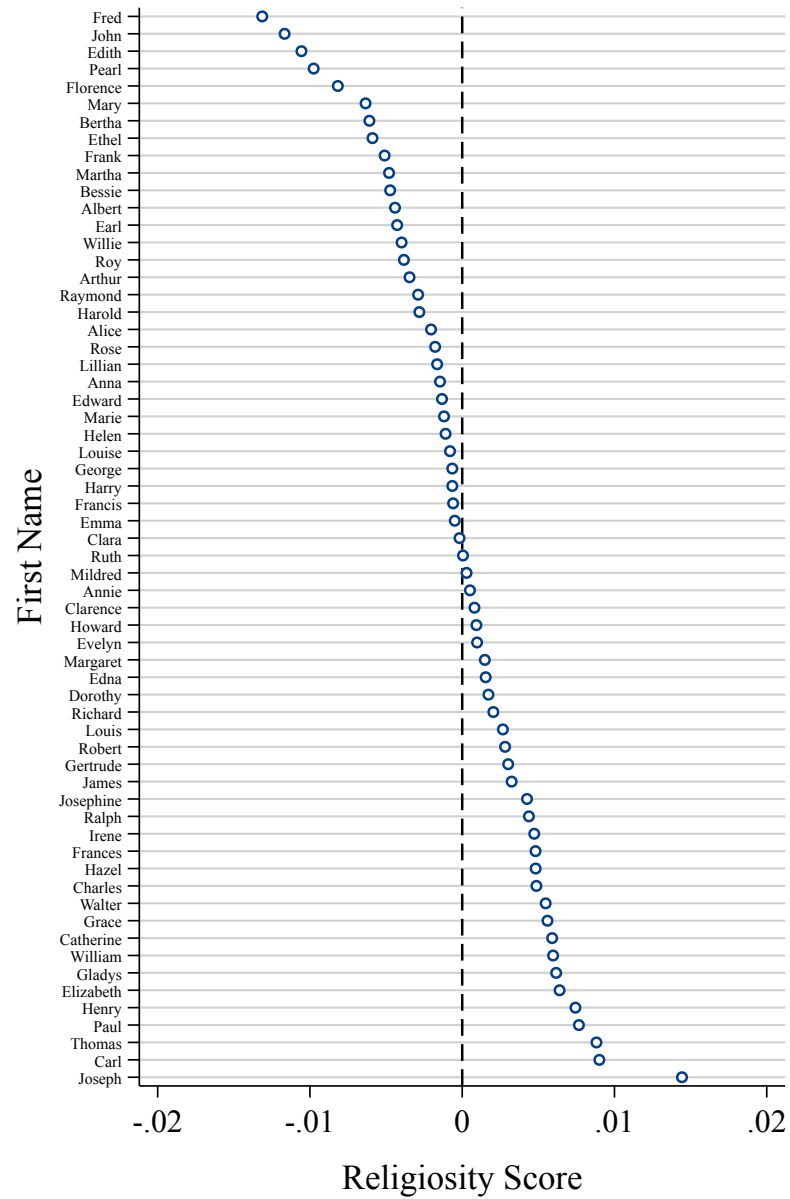
- Boustan, L. P., M. E. Kahn, P. W. Rhode, and M. L. Yanguas (2020). The Effect of Natural Disasters on Economic Activity in US Counties: A Century of Data. *Journal of Urban Economics* 118, 103257.
- Callaway, B., A. Goodman-Bacon, and P. H. Sant'Anna (2021). Difference-in-Differences with a Continuous Treatment. *Working Paper*.
- Callaway, B. and P. H. Sant'Anna (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics* 225(2), 200–230.
- Clay, K., J. Lewis, and E. Severnini (2019). What Explains Cross-City Variation in Mortality During the 1918 Influenza Pandemic? Evidence from 438 US Cities. *Economics & Human Biology* 35, 42–50.
- Clemens, J. and P. Rogers (2023). Demand Shocks, Procurement Policies, and the Nature of Medical Innovation: Evidence from Wartime Prosthetic Device Patents. *Review of Economics and Statistics*, 1–45.
- Crosby, A. W. (1989). *America's Forgotten Pandemic: The Influenza of 1918*. Cambridge: Cambridge University Press.
- Deming, D. (2010). *Science and Technology in World History: Early Christianity, the Rise of Islam and the Middle Ages: 2*. Jefferson, NC: McFarland.
- Eckert, F., A. Gvartz, J. Liang, and M. Peters (2018). A Consistent County-Level Crosswalk for US Spatial Data since 1790. *NBER Working Paper* (No. w26632).
- Esteban, J., L. Mayoral, and D. Ray (2012). Ethnicity and Conflict: An Empirical Study. *American Economic Review* 102(4), 1310–1342.
- Ferrara, A. and P. Fishback (2024). Discrimination, Migration, and Economic Outcomes: Evidence from World War I. *Review of Economics and Statistics* 106(5), 1201–1219.
- Fouka, V. (2019). How do Immigrants Respond to Discrimination? The Case of Germans in the US during World War I. *American Political Science Review* 113(2), 405–422.
- Fouka, V. (2020). Backlash: The Unintended Effects of Language Prohibition in US Schools after World War I. *The Review of Economic Studies* 87(1), 204–239.
- Frost, P. (2020). An Accelerant of Social Change? The Spanish Flu of 1918-19. *International Political Anthropology Journal* 13(2), 123–133.
- Furman, J. L., M. Nagler, and M. Watzinger (2021, November). Disclosure and Subsequent Innovation: Evidence from the Patent Depository Library Program. *American Economic Journal: Economic Policy* 13(4), 239–70.
- Gentzkow, M. and J. M. Shapiro (2011). Ideological Segregation Online and Offline. *The Quarterly Journal of Economics* 126(4), 1799–1839.
- Giuliano, P. and A. Spilimbergo (2023). Recessions, Lifetime Experiences and the Formation of Political Beliefs. *Working paper*.

- Gross, D. P. and B. N. Sampat (2021). The Economics of Crisis Innovation Policy: A Historical Perspective. *American Economic Association: Papers and Proceedings 111*, 346–50.
- Hall, B. H., A. B. Jaffe, and M. Trajtenberg (2001). The NBER Patent Citation Data File: Lessons, Insights and Methodological Tools.
- Hanlon, W. W. (2015). Necessity is the Mother of Invention: Input Supplies and Directed Technical Change. *Econometrica* 83(1), 67–100.
- Huntington, E. (1923). Causes of Geographical Variation in the Influenza Epidemic in the Cities of the United States. *Bull. Nat. Res. Council* 6, 1–36.
- Iannaccone, L. (1998). Introduction to the Economics of Religion. *Journal of Economic Literature* 36(3), 1465–1496.
- Iyer, S. (2016). The New Economics of Religion. *Journal of Economic Literature* 54(2), 395–441.
- Jurajda, Š. and D. Kovač (2021). Names and Behavior in a War. *Journal of Population Economics* 34(1), 1–33.
- Kelly, B., D. Papanikolaou, A. Seru, and M. Taddy (2021). Measuring Technological Innovation Over the Long Run. *American Economic Review: Insights* 3(3), 303–20.
- Lecce, G., L. Ogliari, and M. P. Squicciarini (2021). Birth and Migration of Scientists: Does Religiosity Matter? Evidence from 19th-Century France. *Journal of Economic Behavior & Organization* 187, 274–289.
- Long, J. and J. Ferrie (2013). Intergenerational Occupational Mobility in Great Britain and the United States since 1850. *American Economic Review* 103(4), 1109–1137.
- Markel, H., H. B. Lipman, J. A. Navarro, A. Sloan, J. R. Michalsen, A. M. Stern, and M. S. Cetron (2007). Nonpharmaceutical Interventions Implemented by US Cities During the 1918-1919 Influenza Pandemic. *Journal of the American Medical Association* 298(6), 644–654.
- Martin, G. J. and A. Yurukoglu (2017). Bias in Cable News: Persuasion and Polarization. *American Economic Review* 107(9), 2565–2599.
- Miao, Q. and D. Popp (2014). Necessity as the Mother of Invention: Innovative Responses to Natural Disasters. *Journal of Environmental Economics and Management* 68(2), 280–295.
- Mokyr, J. (2011). The Economics of Being Jewish. *Critical Review* 23(1-2), 195–206.
- Montalvo, J. G. and M. Reynal-Querol (2005). Ethnic Polarization, Potential Conflict, and Civil Wars. *American economic review* 95(3), 796–816.
- Montgomery, L. M. (1992[1924]). In M. Rubio and E. Waterston (Eds.), *The Selected Journals of L.M. Montgomery, Volume III: 1921-1929*. Toronto: Oxford University Press.
- Moscona, J. (2022). Environmental Catastrophe and the Direction of Invention: Evidence from the American Dust Bowl. *Working Paper*.
- Norenzayan, A. and I. G. Hansen (2006). Belief in Supernatural Agents in the Face of Death. *Person-*

- ality and Social Psychology Bulletin* 32(2), 174–187.
- Olivetti, C., M. D. Paserman, L. Salisbury, and E. A. Weber (2020). Who Married, (to) Whom, and Where? Trends in Marriage in the United States, 1850–1940. *NBER Working Paper* 28033.
- Pargament, K. I. (2001). *The Psychology of Religion and Coping: Theory, Research, Practice*. New York: Guilford Press.
- Phillips, H. (2020). ‘17, ‘18, ‘19: Religion and Science in three Pandemics, 1817, 1918, and 2019. *Journal of Global History* 15(3), 434–443.
- Posch, M. (2022). Do Disasters Affect the Tightness of Social Norms?
- Rambachan, A. and J. Roth (2023). A More Credible Approach to Parallel Trends. *Review of Economic Studies* 90(5), 2555–2591.
- Ruggles, S., S. Flood, S. Foster, R. Goeken, J. Pacas, M. Schouweiler, and M. Sobek (2021). IPUMS USA: Version 11.0 [dataset]. [Machine-readable database].
- Sibley, C. G. and J. Bulbulia (2012). Faith After an Earthquake: A Longitudinal Study of Religion and Perceived Health Before and After the 2011 Christchurch New Zealand Earthquake. *PloS One* 7(12).
- Spinney, L. (2018). *Pale Rider: the Spanish Flu of 1918 and How It Changed the World*. London: Vintage.
- Squicciarini, M. P. (2020). Devotion and Development: Religiosity, Education, and Economic Progress in 19th-Century France. *American Economic Review* (110), 3454–3491.
- Stark, R. (1992). The Reliability of Historical United States Census Data on Religion. *Sociological Analysis* 53(1), 91–95.
- Stark, R. (1998). The Rise and Fall of Christian Science. *Journal of Contemporary Religion* 13(2), 189–214.
- Sun, L. and S. Abraham (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics* 225(2), 175–199.
- Weber, M. (1905). *The Protestant Ethic and the Spirit of Capitalism*. London: Routledge.

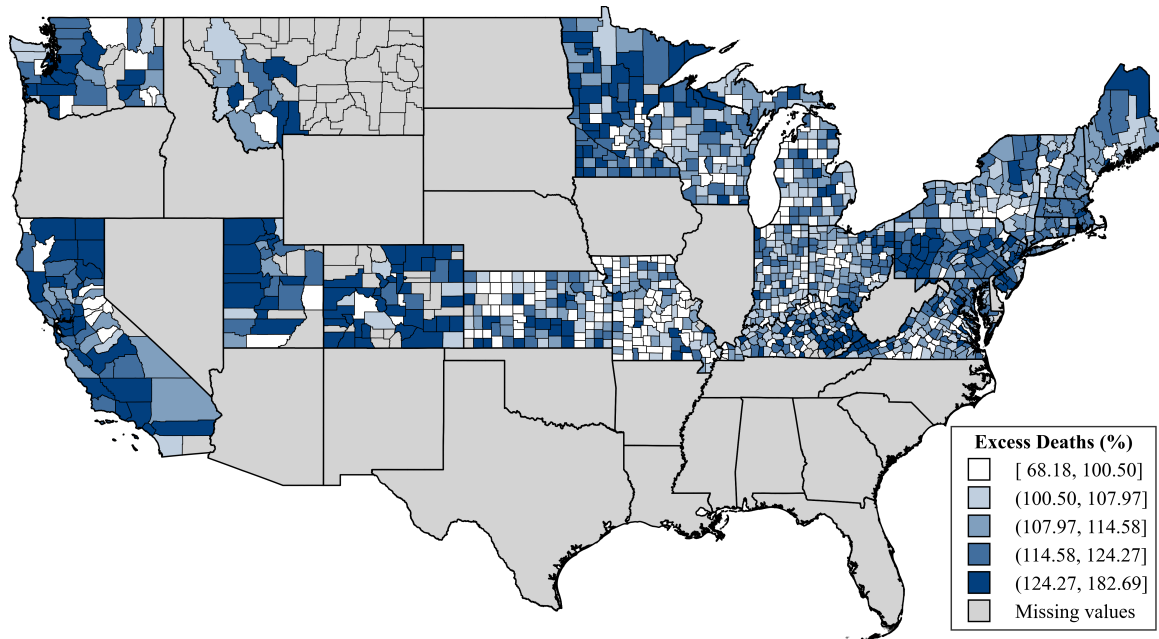
Figures

Figure 1: Name-Based Religiosity Scores



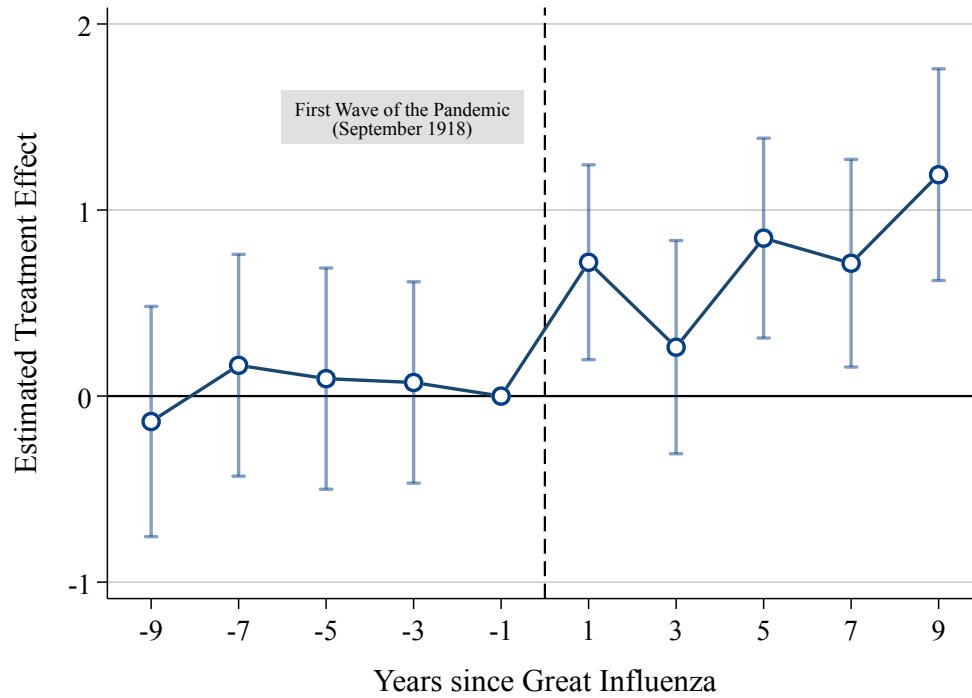
Notes: This figure displays the religiosity scores estimated using equation (1). The sample includes data from the 1906–1916 Censuses of Religious Bodies and from the U.S. Census on individuals born in the United States between 1896 and 1916. We estimate religiosity scores for names appearing in at least 0.3% of the sample. Coefficients are reported in increasing order.

Figure 2: Spatial Distribution of Excess Mortality During the Great Influenza Pandemic



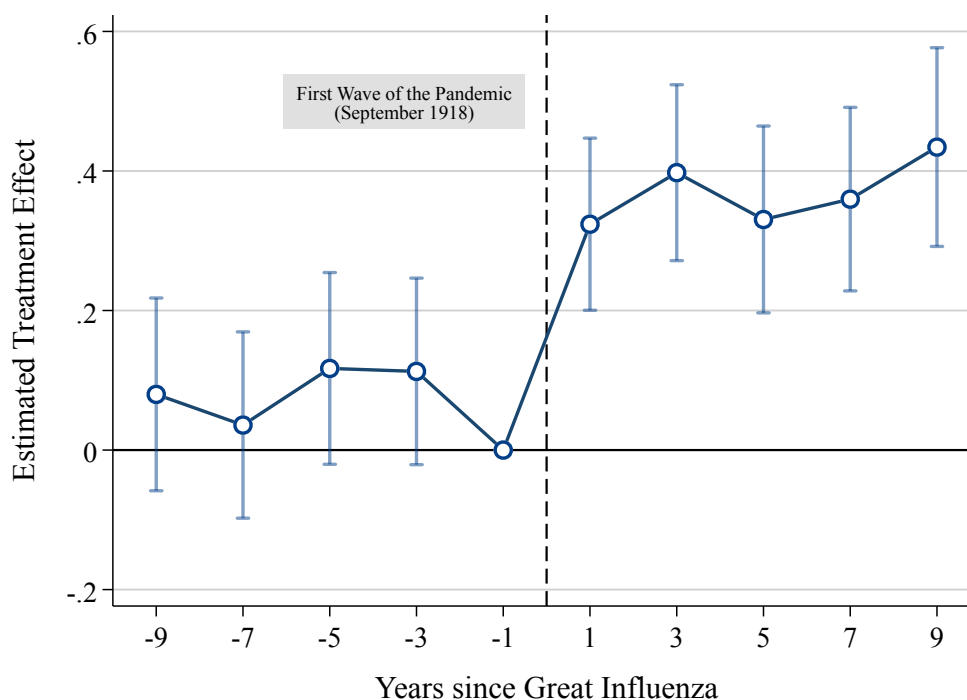
Notes: This figure displays geographic variation in excess deaths, defined in equation (3). Excess mortality is the ratio between the average yearly number of deaths during the pandemic years (1918–1919) and the average yearly number of deaths in the three years before the pandemic (1915–1917). Mortality statistics before 1915 are not available. Excess mortality is shown as a percentage. Lighter-to-darker blue indicates increasing excess deaths. Counties are displayed at their 1920 borders. Mortality data are unavailable for the states and counties shown in gray, and these are therefore excluded from the analysis.

Figure 3: Impact of the Pandemic on Religiosity



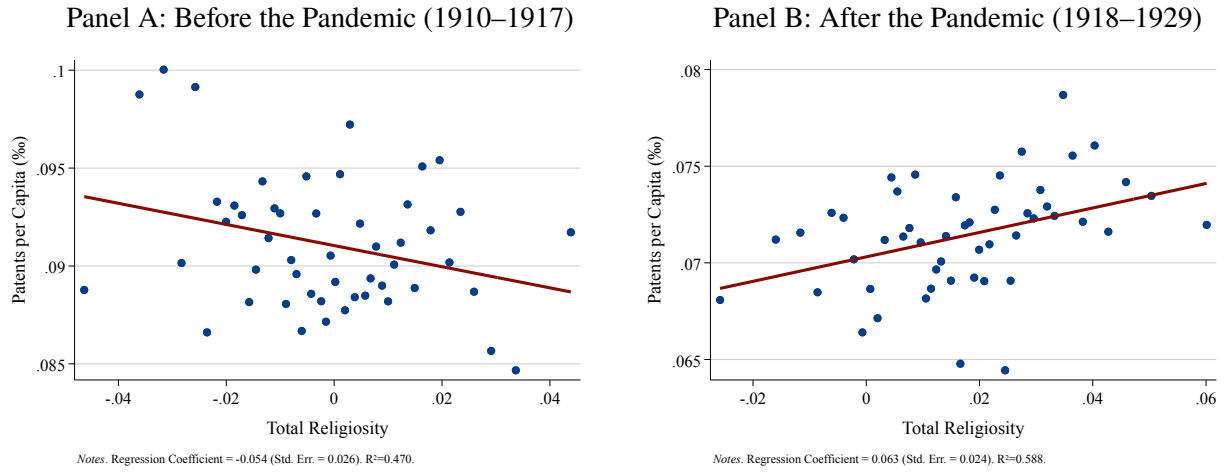
Notes: This figure displays the dynamic treatment effects of the pandemic on overall religiosity. The unit of observation is a county observed at a biennial frequency. Each dot reports the coefficient of an interaction between the baseline measure of excess deaths, defined in equation (3), and a biennial time dummy. The coefficient for 1917–1918 serves as the baseline category. The specification includes county and state-by-year fixed effects and controls for an interaction term between the population in 1910 and a post-treatment indicator. Bands report 90% confidence intervals. Standard errors are clustered at the county level. The dashed vertical line indicates the timing of the first wave of the pandemic (September 1918).

Figure 4: Impact of the Pandemic on Innovation



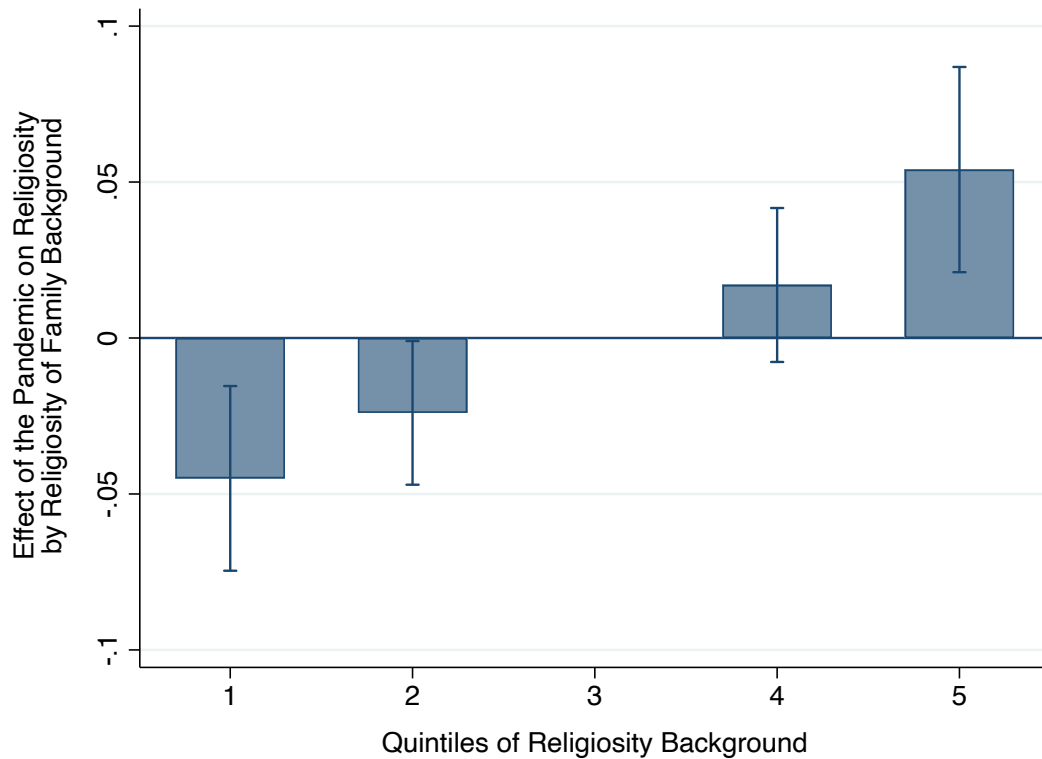
Notes: This figure displays the dynamic treatment effects of the pandemic on innovation. The dependent variable is the $\log(1 + \text{total number of patents})$ filed in a given year. The unit of observation is a county observed at a biennial frequency. Each dot reports the coefficient of an interaction between the baseline measure of excess deaths and a biennial time dummy. The coefficient for the biennial 1917–1918 serves as the baseline category. The model includes county- and state-by-year fixed effects and controls for the interaction between the 1910 population and a post-treatment indicator. Bands report 90% confidence intervals. Standard errors are clustered at the county level. The dashed vertical line indicates the timing of the first wave of the pandemic (September 1918).

Figure 5: Correlation Between Religiosity and Science



Notes: These figures display county-level binned scatter plots reporting the correlation between science (measured as patenting activity divided by population) and religiosity. Patents per capita are winsorized at the top and bottom 1%. The unit of observation is a county observed at yearly frequency. Religiosity is defined as described in section 3.1. The estimates in both plots are residualized and shown after accounting for county and year fixed effects. We report regression coefficients with standard errors and associated R^2 separately for each plot.

Figure 6: Impact of the Pandemic on the Polarization of Religious Beliefs



Notes: This figure displays the estimated impact of the pandemic on the polarization of religious beliefs by quintiles of background religiosity. Each bar reports the coefficient of an interaction between the baseline measure of excess deaths, a post-treatment indicator, and an indicator for the quintile of background religiosity. The unit of observation is a child born between 1900 and 1929. Treated children are those born after the influenza, i.e., after 1918. The dependent variable is the religiosity score associated with the child's name. Background religiosity is measured as the religiosity score of the child's father. The third quintile of background religiosity serves as the baseline category. The specification includes individual fixed effects at the household-head level: race, urban status, and place of birth. It also includes county-by-cohort, county-by-quintile of religious background, and cohort-by-quintile of religious background fixed effects. Standard errors are clustered at the county level, and the bands report the 90% confidence interval for each coefficient.

Tables

Table 1: Impact of the Pandemic on Religiosity

	Share of Affiliated			Name-Based Religiosity		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Catholics	Protestants	All	Catholics	Protestants
Excess Deaths $\times$ Post	0.220*** (0.025)	0.078*** (0.013)	0.094*** (0.017)	0.863*** (0.199)	0.269** (0.111)	0.688*** (0.171)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	–	–	–
State-Year FE	–	–	–	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	3,825	3,825	3,825	38,220	38,220	38,220
R ²	0.851	0.904	0.929	0.648	0.522	0.678

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity. The unit of observation is a county observed at a decade frequency between 1906 and 1926 (in columns 1–3) and a yearly frequency between 1900 and 1929 (in columns 4–6). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–3), the dependent variable is the number of individuals affiliated with religious denominations, as enumerated in the Census of Religious Bodies, divided by the 1900 county population; in columns (4–6), the dependent variable is the name-based religiosity measure described in the main text. Columns (1) and (4) report the effect of the pandemic on overall religiosity. Columns (2) and (5) report the effect of the pandemic on Catholicism. Columns (3) and (6) report the effect of the pandemic on Protestantism. All columns include county fixed effects and control for an interaction term between population in 1910 and a post-treatment indicator. Columns (1) to (3) include state-by-decade fixed effects, while columns (4) to (6) include state-by-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Impact of the Pandemic on Employment in Scientific Occupations

	Whole Population			Skilled Population		
	(1)	(2)	(3)	(4)	(5)	(6)
	Share STEM	Share Medical	Share Non-Medical	Share STEM	Share Medical	Share Non-Medical
Excess Deaths $\times$ Post	0.934*** (0.108)	0.331*** (0.046)	0.604*** (0.081)	15.374*** (3.070)	5.310*** (1.179)	10.064*** (2.134)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	5,098	5,098	5,098	5,096	5,096	5,096
R ²	0.733	0.687	0.744	0.548	0.575	0.565

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at a decade frequency between 1900 and 1930. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. All specifications include county fixed effects, state-by-decade fixed effects, and control for an interaction term between population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Impact of the Pandemic on Innovation

	Overall Effect			Intensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
	All Patents	Pharmaceutical	Non-Pharmaceutical	All Patents	Pharmaceutical	Non-Pharmaceutical
Excess Deaths $\times$ Post	0.370*** (0.056)	0.236*** (0.041)	0.319*** (0.051)	0.650*** (0.210)	0.711*** (0.224)	0.593*** (0.208)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	265	265	265
Observations	38,250	38,250	38,250	7,950	7,950	7,950
R ²	0.858	0.791	0.852	0.898	0.801	0.888

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the log(1+ number of granted patents). In columns (2) and (5), the dependent variable is the log(1+ number of granted pharmaceutical patents). In columns (3) and (6), the dependent variable is the log(1+ number of granted non-pharmaceutical patents). In columns (4) to (6), we restrict the sample to counties with at least one granted patent every year. All regressions include county fixed effects, state-by-year fixed effects, and a control for the interaction between population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Impact of the Pandemic on Individual Religiosity: STEM and Non-STEM

	Whole Population	Skilled Population
	(1)	(2)
	Name-Based Religiosity	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ Employed in STEM	-0.061*** (0.023)	-0.061*** (0.023)
Individual Controls	Yes	Yes
County–Year FE	Yes	Yes
County–Employed in STEM FE	Yes	Yes
Year–Employed in STEM FE	Yes	Yes
Number of Counties	1,274	1,274
Observations	10,526,699	3,678,297
R ²	0.008	0.014

Notes: This table displays the impact of exposure to the pandemic on the religiosity of individuals employed in STEM occupations or in other occupations. In column (1), the unit of observation is a child born between 1900 and 1929. In column (2), the unit of observation is a child born between 1900 and 1929 in a household where the household head is employed in a skilled occupation. Religiosity is defined as the religiosity score associated with the child’s name. “Post” is a categorical variable equal to zero for children born before the pandemic—i.e., before 1918—and one for those born after the pandemic—i.e., in or after 1918. The baseline treatment “Excess Deaths” is defined in equation (3). “Employed in STEM” is an indicator variable equal to one if the household head is employed in a STEM occupation or zero otherwise. In all specifications, the observations are weighted by the inverse of the number of children in the household. All specifications include a vector of individual controls for the household-head level: race, urban status, and place of birth, as well as STEM-by-county, STEM-by-cohort, and county-by-cohort fixed effects. The sample includes the entire population in column (1) and only individuals employed in skilled occupations—as defined in Table B.2—in column (2). Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Impact of the Pandemic by Religious Background

	Whole Population		Skilled Population	
	(1)	(2)	(3)	(4)
	Employed in STEM	Name-Based Religiosity	Employed in STEM	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ High Religious Background	-0.701** (0.337)	0.058*** (0.012)	-1.558* (0.849)	0.045** (0.020)
Individual Controls	Yes	Yes	Yes	Yes
County–Year FE	Yes	Yes	Yes	Yes
High Religious Background–Year FE	Yes	Yes	Yes	Yes
High Religious Background–County FE	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,275
Observations	5,347,751	4,246,353	2,229,754	1,566,801
R ²	0.016	0.058	0.030	0.080

Notes: This table displays the impact of exposure to the pandemic on occupational choice—columns (1) and (3)—and on religiosity—columns (2) and (4)—by individual-level background religiosity. The unit of observation in columns (1) and (3) is an adult. The dependent variable “Employed in STEM” is an indicator equal to one if an individual is employed in a STEM occupation—as defined in Table B.2—or zero otherwise. “Post” is an indicator equal to one if an individual is aged between 18 and 25 in 1918 or zero otherwise. In columns (2) and (4), the unit of observation is a head of household observed once for each child born between 1900 and 1929. “Name-Based Religiosity” is defined as the religiosity score associated with the child’s name. The observations are weighted by the inverse of the number of children in the household. “Post” is an indicator equal to one for children born during and after the pandemic—i.e., over 1918–1929—or zero for those born before the pandemic—i.e., before 1918. The baseline treatment “Excess Deaths” is defined in equation (3). “High Background Religiosity” is an indicator variable taking value one if the religiosity score of the name of the head of the household is equal to or above the median, or zero if it is below the median. The table displays the coefficient of the interaction between these terms. All specifications include a vector of individual controls for the household head: race, urban status, and place of birth, as well as county-by-year, high religious background-by-year, and high religious background-by-county fixed effects. The sample includes the entire population in columns (1)–(2) and only individuals employed in skilled occupations—as defined in Table B.2—in columns (3)–(4). Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Online Appendix

Dealing with Adversity: Religiosity or Science? Evidence from the Great Influenza Pandemic

Enrico Berkes, Davide M. Coluccia, Gaia Dossi, and Mara P. Squicciarini

May, 2026

A Data Appendix

This section lists the data sources and describes how we construct the variables used in the analysis.

A.1 Names

Data on names are from the individual full-count US 1930 population census (Ruggles et al., 2021). Since first name records in the census conflate first and middle names, we consider only the first word that appears in each string. The baseline analysis includes all names that appear in at least 0.3% of the population of US-born children between 1900 and 1930. We perform a robustness exercise and show that the baseline results are not sensitive to alternative frequency thresholds. We exclude first-generation immigrants because their names would not reflect exposure to the pandemic, as they were not born in the United States.

A.2 Religious Affiliations

Data on religious affiliations are available from the National Historical Geographic Information System (NHGIS) and originate from the Census of Religious Bodies, which was administered at 10-year intervals between 1906 and 1936. We discard the 1936 census because prior research indicates that uptake was low and uneven across counties (Stark, 1992). Census enumerators asked churches, congregations, and other local organizations to report the number of affiliates. The data was then aggregated at the county level. In our analysis, “Total Religiosity” is computed as the sum of religious members across all denominations; “Catholics” are enumerated as such. We collectively refer to “Protestants” as a set of denominations that we manually mapped to some branch of Protestantism (including, e.g., the Methodist, Evangelical, and various Baptist churches).

We use data on the names of saints and biblical characters from Abramitzky et al. (2016) to develop an additional indicator of county-level religiosity. To construct this additional measure, we use the baseline equation (2) and substitute the estimated $\hat{\gamma}^k$ coefficients with three indicator variables equal to one if the given name is that of a saint, a biblical character, or either.

A.3 Patents

A.3.1 Patent Data

Patent data are from Berkes (2018), who performed optical character recognition (OCR) on original patent documents issued by the United States Patents and Trademark Office between 1836 and 2010. Information includes the filing and issue year, author name, latitude and longitude of the inventor(s), and inferred USPC technology class. The data contain a set of additional variables, including the complete text of the patent document and the issue year of the patent, not used in our analysis. We geo-code each patent to its 1920 county using boundary shapefiles from NHGIS. When we collapse by

county-year, we weight each patent by the inverse of the number of technology classes and the inverse of the number of authors. Hence, a patent with two authors and two technological classes appears four times in the original patent-level dataset, and each instance is assigned a .25 weight when aggregating at the county level. We code USPC classes to the NBER classification (Hall et al., 2001). We modify the canonical NBER classification and conflate the “Chemical” and “Drugs” categories into a single “Pharmaceuticals” class. Since multiple USPC codes are typically assigned to a single patent, most patents that fall under “Drugs” would also appear as “Chemical.” To avoid this, we recast them into one category. It is worth noting that all the results we present regarding pharmaceutical patents also hold if we keep the “Chemical” and “Drugs” classes separate.

A.3.2 Patent Importance

We measure patent ‘importance’ using the measure developed by Kelly et al. (2021). From their data, we derive two metrics. One is the number of “Breakthrough” innovations, which are defined as any patent whose importance is in the top 20% of the overall importance distribution. The second variable is the share of breakthrough patents relative to the overall number of issued patents. Both measures are net of grant-year fixed effects. We use forward and backward similarity defined within a 5-year window around the issue year of the patent.

A.4 Occupational Structure

We use the 1950 harmonized occupation classification (OCC1950) and manually map occupational codes to STEM and medical occupations as described in Table B.2.

A.5 Controls & Mortality Statistics

We extract individual-level information on race and urban-rural status from the IPUMS full-count data.

County-level covariates are from NHGIS. This aggregates individual-level data from population censuses, manufacturing, and agricultural censuses. All data come at historical county borders.

Mortality statistics are likewise provided by NHGIS. For the period we are interested in, 1915-1919, they were collected for about 1,200 counties, covering approximately 60% of the US population. We measure Influenza-related mortality as the ratio of average yearly deaths in 1918 and 1919 to yearly deaths in 1915 to 1917.⁴⁷

A.6 Other Data

In several robustness regressions, we control for WW1 mortality. The underlying data are from Ferrara and Fishback (2024) and were kindly shared by the authors.

⁴⁷The original documents report, for major cities, deaths broken down by (alleged) cause. We do not use this data for two main reasons. First, they are incomplete and are only available for cities. Second, Beach et al. (2022) criticizes the methodology used to impute the causes of death.

A.7 Boundary Harmonization

County-level data from NHGIS and other sources are typically provided at historical borders. To ensure comparability and consistency, we adopt the methodology developed by Eckert, Gvartz, Liang, and Peters (2018) to compute geographical crosswalks between US counties over time. In a nutshell, their methodology is as follows. Suppose we know the distribution of a given variable y across counties at decade frequency between 1900 and 1930. To harmonize borders over one year, Eckert et al. (2018) overlay the shapefile of counties in a given year, say, 1900, to that in the reference year, say, 1920. They then compute the percentage of land that a given county shares with itself and with other counties between the two years, and that is assigned to other counties. To construct the harmonized variable, one multiplies these overlapping-area weights by the variable recorded in 1900 and aggregates them across 1920-border counties. The underlying assumption is that y is evenly distributed over the county territory. While this assumption may seem too demanding in some cases, departures from it are plausibly innocuous in our setting. First, county borders had undergone major consolidations before 1900 and remained stable thereafter. Moreover, mortality data are mostly available for the Northeast and Midwest areas, where boundary changes were rare and minor after the 1890s. In our analysis, we map all county-level variables to 1920 borders.

A.8 Details on Sample Construction

In this paragraph, we provide additional technical details on how we construct the estimation samples. The main sample restriction comes from the fact that we observe mortality in 1,302 out of 3,100 counties. We then discard 27 counties with values of excess mortality below or above, respectively, the bottom 1% (85% of the pre-pandemic mortality) and top 1% (180% of the pre-pandemic level) of the excess-mortality distribution. Because such figures are due to scarcely-inhabited areas, these 27 counties account for less than 0.1% of the population in the 1,302-county sample. We are left with 1,275 counties. In the rest of the paragraph, we explain why we may not always be able to use all 1,275 for the estimation.

A.8.1 *County-Level Religiosity*

The yearly county religiosity estimation sample is a balanced panel in which each county is observed at yearly intervals between 1900 and 1929. This implies that the number of counties in this balanced panel may not be 1,275 if at least one county is not observed at least once between 1900 and 1929. This happens because, especially in sparsely inhabited areas, our name-frequency threshold may mean we cannot match any newborn in a given cohort. If that is the case, the county will subsequently be dropped from the estimation sample. This happens for one county, so the estimation sample for the county-level religiosity analysis consists of 1,274 counties accounting for 99.5% of the population in

the 1,302-county sample.

In the decade-level county panel, counties are observed at decade frequency instead. Religiosity is constructed from the Census of Religious Bodies, as the number of people affiliated with a denomination, normalized by the 1910 population, i.e., the last year before the Influenza. In this case, the post-treatment indicator returns a value of one for the decades 1920 and 1930 and zero otherwise.

A.8.2 County-Level Innovation

The yearly county-level innovation sample is a balanced panel in which each county is observed at yearly intervals between 1900 and 1929. Thus, an observation in the dataset can be either a number greater than zero (if one or more patents are observed in that county-year) or zero (if no patents are observed). The estimation sample in this case includes all 1,275 counties for which we observe mortality. When looking at the intensive margin, the sample is a balanced panel in which each county with at least one patent in any year is observed annually between 1900 and 1929. The estimation sample in this case is reduced to 265 counties.

The decade-county-level innovation sample is a balanced panel in which each unit is observed at a decade frequency. In this case, the sample is constructed from adults observed once per census decade between 1900 and 1930. The post-treatment variable is an indicator equal to one for the decades 1920 and 1930 and zero otherwise. The dependent variable is the share of individuals employed in a STEM occupation (listed in Table B.2), normalized by the 1910 population, i.e., the last census before the Influenza.

A.8.3 Individual-Level

We construct two individual-level datasets. The first sample (“adult sample”) comprises all those born before 1900. Each individual is observed once in the 1930 census. Because the adult sample is used to study the evolution of the occupational structure, we discard (i) individuals with missing occupational response, (ii) farmers, who have been shown to display disproportionately high occupational persistence (Long and Ferrie, 2013), and (iii) women, since female labor force participation was extremely low. The second sample (“children sample”) comprises all those born between 1900 and 1930. Each household is observed once per child. In the children sample, we excluded households when (i) the father’s occupation was missing and (ii) there was no father, since here, too, the labor force participation of women was very low. We identify a household as “STEM” if at least one of its members is employed in a STEM occupation. In addition, we excluded a small fraction of households ($\approx 0.7\%$) in which we could not unambiguously identify the father. Consistent with the county-level analysis, we do not assign a religiosity score to first-generation immigrants.

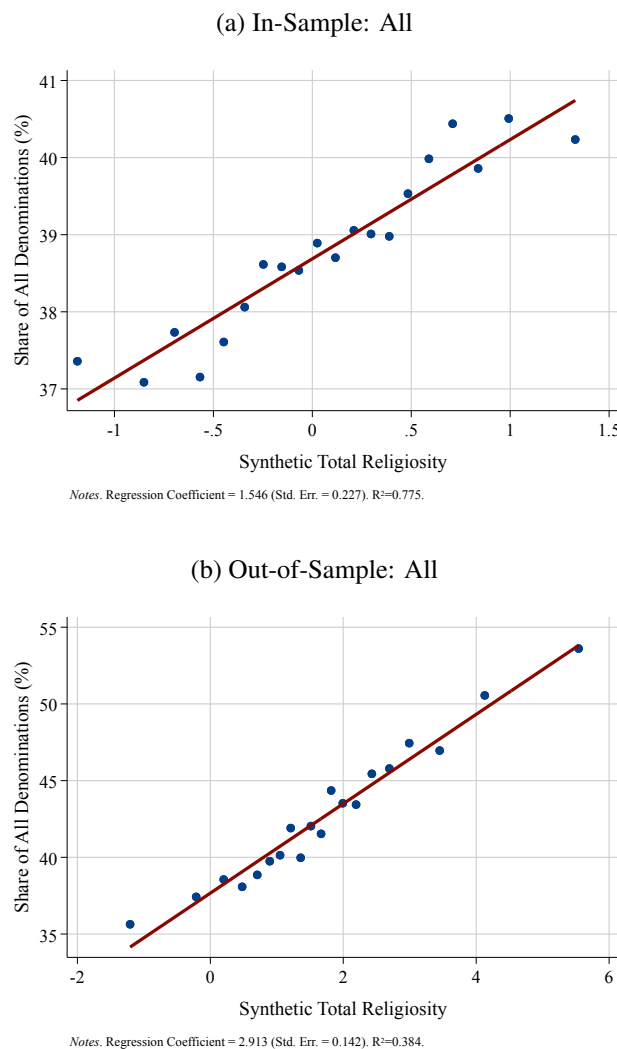
A.8.4 City-Level

To build the city-level sample, we construct the baseline excess deaths treatment variable from data by Clay et al. (2019). The dataset contains information for 982 cities. We observe mortality continuously between 1915 and 1919—the period over which we compute excess mortality in the county-level sample—for a subset of 493 cities. Moreover, we do not observe population data for 1900 for four additional cities, and in 44 cases we do not observe newborns in any year of the sample period. The final sample of 489 cities is thus reduced to 445 for religiosity regressions. As in the county-level sample, we exclude cities in the top and bottom 1% of the excess mortality distribution. In Figure B.6, we report the location of each of the 489 cities and the number of cities included in the sample by state. We geocode patents to historical city borders and construct the name-based religiosity measure from individuals recorded as living in each city in the 1930 census. The city-level sample is used in the regressions shown in Table B.9.

B Additional Figures and Tables

B.1 Appendix Figures

Figure B.1: In-Sample and Out-of-Sample Fit of Name-Based Religiosity Measure



Notes: These figures display county-level binned scatter plots reporting the correlation between our religiosity measure and the number of affiliated members to all denominations (B.1a-B.1b) divided by population in 1900. In-sample figures report data for the 1906 and 1916 censuses of religious affiliations. Out-of-sample figures instead report data for 1926. In-sample regressions include county-fixed effects; out-of-sample regressions include state-fixed effects. In the note, we report the regression coefficients and the associated R^2 .

(a) Text

1,619,005

SAMUEL M. STRONG, OF GARDEN CITY, NEW YORK.

RESPIRATION AND PULSE RECORDER.

This invention relates to a device or instrument for recording the character of the actions of the heart and respiratory organs of a person.

3 The primary object of the invention is to
provide an instrument which will produce an
accurate graphic representation of the rate,
rhythm, and force of respiration and pulse
of a human being over a short or a long
10 period of time.

ing plate 13 and a vertical portion 19 adapted to be placed against a side plate of the casing 10. The main bearing plate fits snugly within the casing and one end of the horizontal portion 18 abuts against the cover 11 when the latter is in position. A side bearing plate 20 is located immediately adjacent to the detachable side plate 11 and an intermediate bearing plate 21 is interposed between the bearing plate 20 and the

(b) Figures

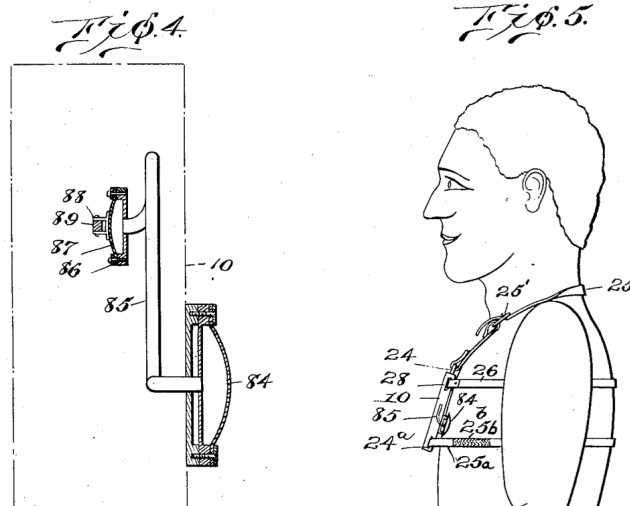
S. M. STRONG

1,619,005

RESPIRATION AND PULSE RECORDER

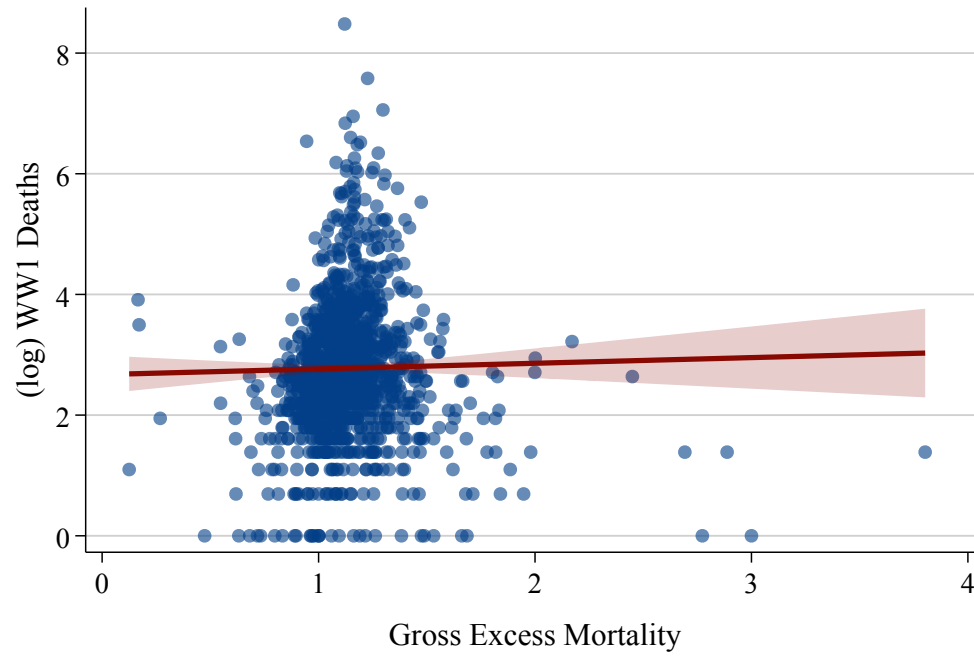
Filed Jan. 11, 1922

2 Sheets-Sheet 2



Notes: This figure displays the text and figures of a patent classified as pharmaceutical.

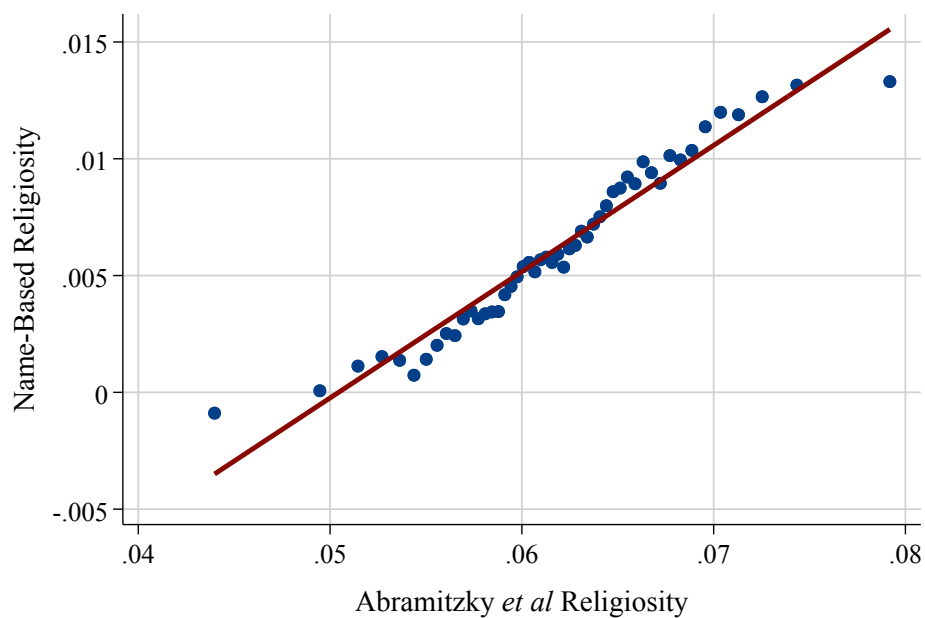
Figure B.3: Correlation Between WW1 Deaths and Excess Deaths



Notes. Regression Coefficient = 0.094 (Std. Err. = 0.140). $R^2=0.000$.

Notes: This figure displays the correlation between WW1 deaths (in log) and excess mortality due to the Influenza pandemic. In the notes, we report the regression coefficient and the R^2 . Data on WW1 deaths are from Ferrara and Fishback (2024).

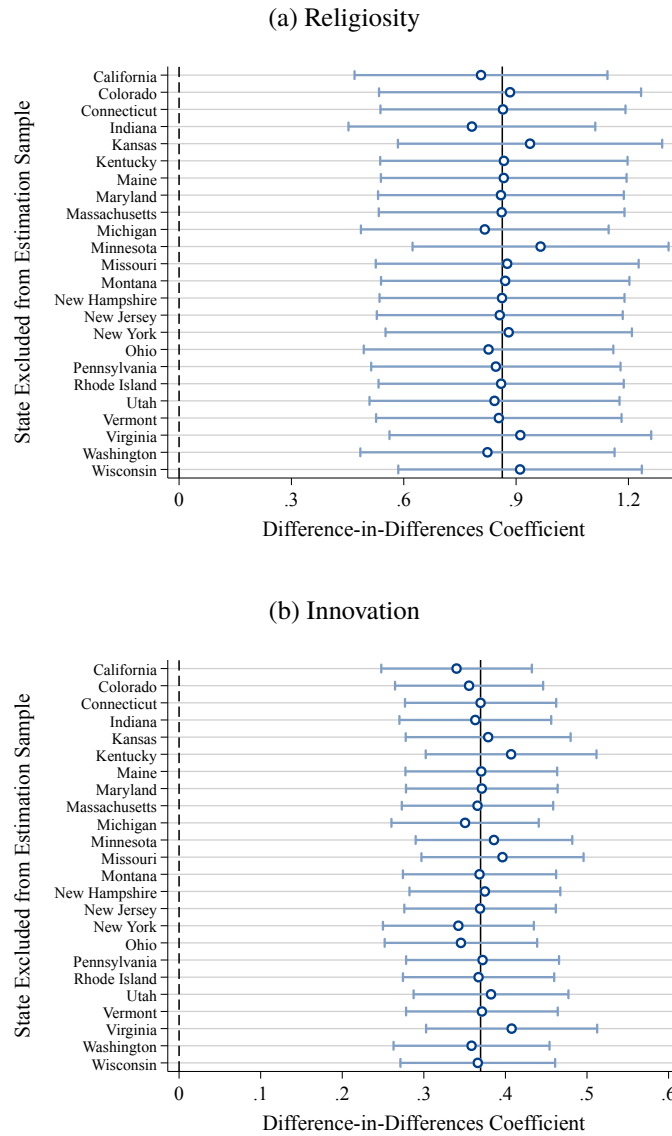
Figure B.4: Correlation Between Abramitzky et al. (2016) Religiosity and Baseline Religiosity



Notes. Regression Coefficient = 0.541 (Std. Err. = 0.018). $R^2=0.584$.

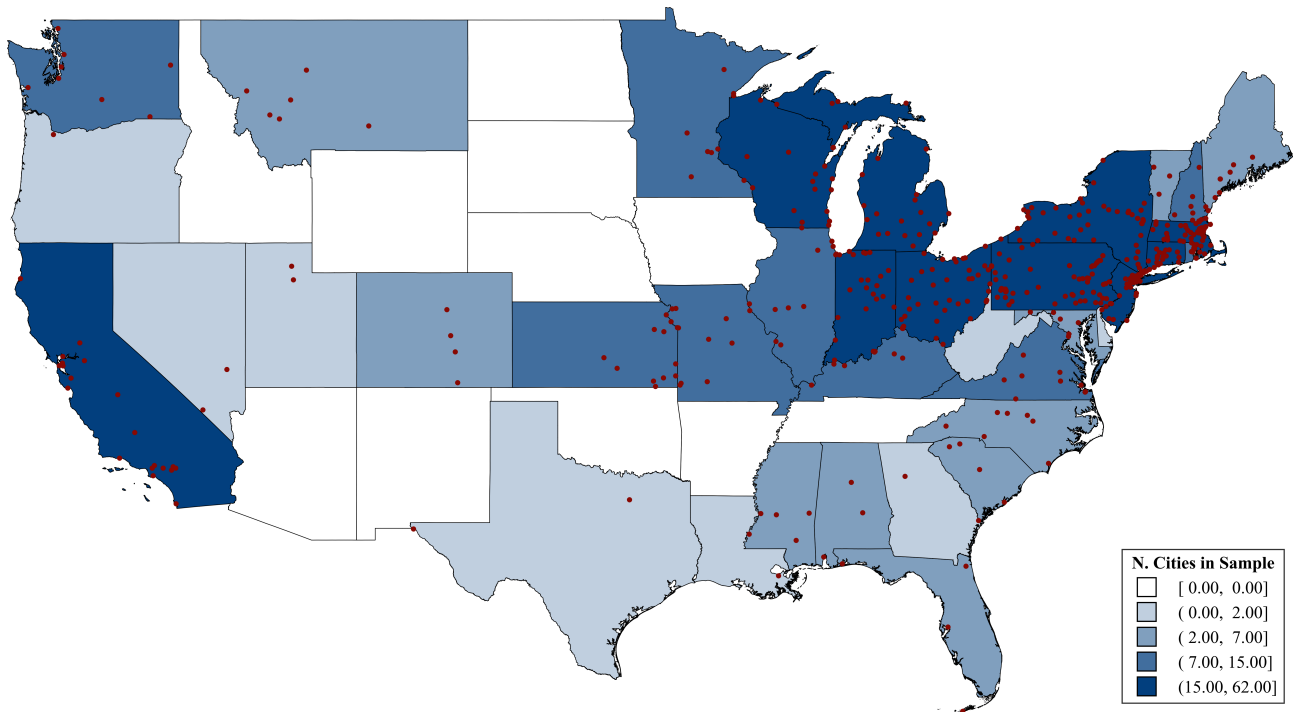
Notes: This figure displays the correlation between our baseline religiosity measure (multiplied by 100) and the share of biblical or saint names, as defined in Abramitzky et al. (2016). The unit of observation is a county observed at yearly frequency between 1900 and 1930. The graph partials out county and year fixed effects. The notes report the regression coefficient and the associated standard error, clustered at the county level, and R^2 .

Figure B.5: County-Level Estimates Excluding One State at a Time



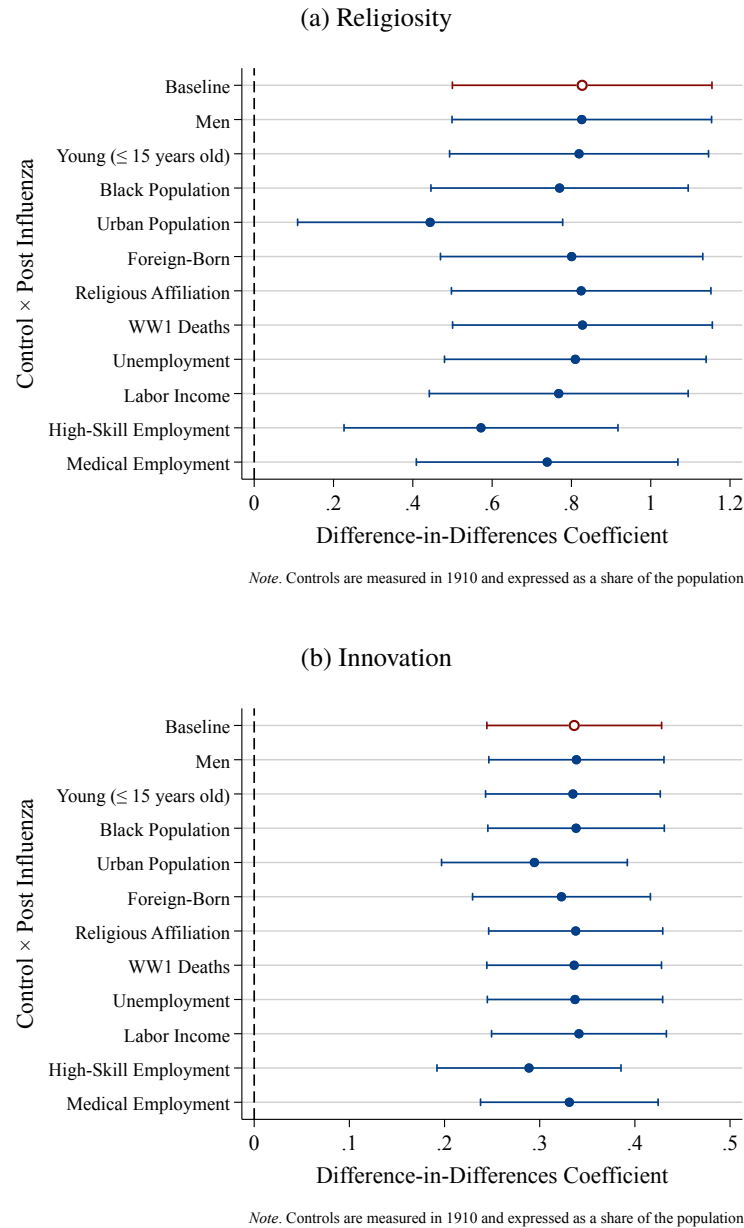
Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity and innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. Each dot reports the coefficient of an interaction between a “Post” categorical variable equal to one during and after the pandemic—i.e., in or after 1918—or zero otherwise and the baseline treatment “Excess Deaths” defined in equation (3), dropping one state at a time from the estimation sample. In Panel B.5a, the dependent variable is the name-based religiosity measure; in Panel B.5b, the dependent variable is the log(1+ number of granted patents). All regressions include county- and state-by-year fixed effects and control for the interaction between population in 1910 and a post-treatment indicator. Standard errors are clustered at the county level, and the bands report 90% confidence intervals. The solid black line reports the treatment effect on the full sample.

Figure B.6: Distribution of Cities in the City-level Sample



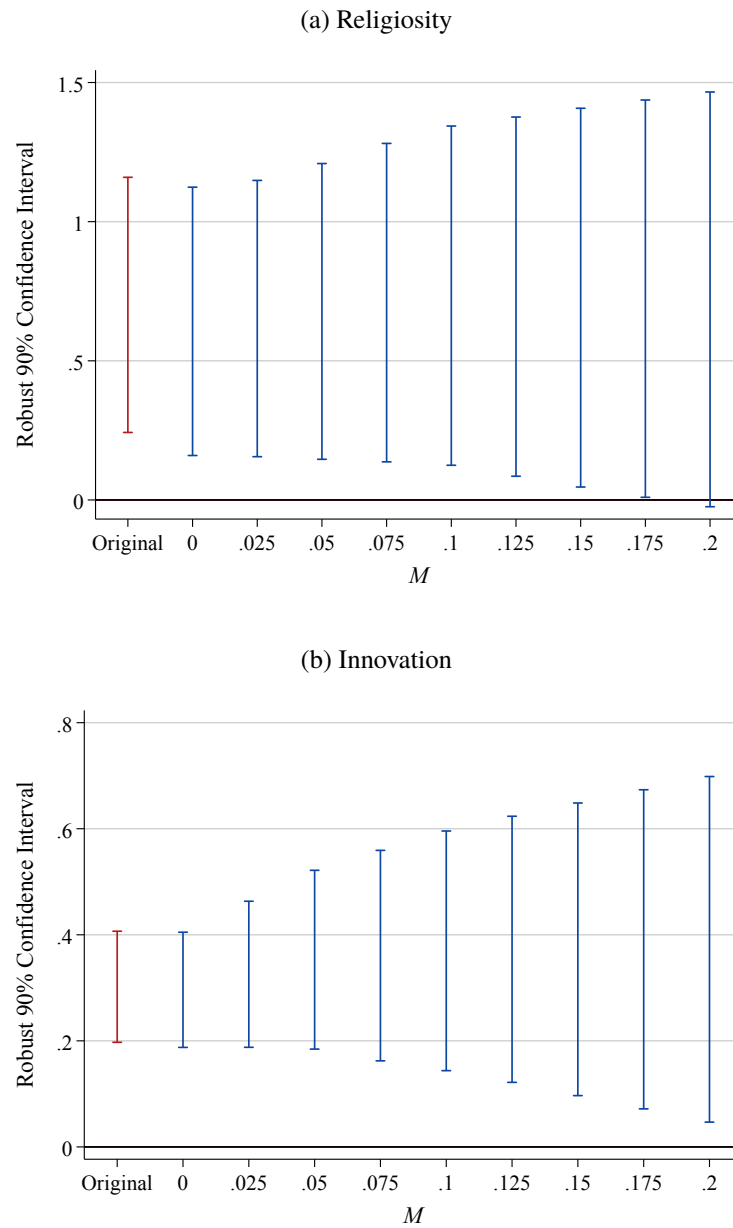
Notes: This figure displays the spatial distribution of the cities in the city-level sample used in Table B.9. We use data from Clay et al. (2019). The red dots report the coordinates of the 489 cities for which we can construct the excess mortality treatment measure. Lighter to darker shades of blue indicate the state-level number of cities included in the final sample.

Figure B.7: Robustness of Religiosity and Innovation Response to Observed Heterogeneity



Notes: This figure displays sensitivity analyses of the estimated effect of the pandemic on religiosity (Panel B.7a) and innovation (Panel B.7b) to pre-treatment differences in observable variables. Each panel plots the baseline coefficient from regression (4) in red. The blue dots show the corresponding estimates obtained when adding one control at a time, each interacted with the post-Influenza indicator. All controls are measured in 1910—the last census before the pandemic—and are expressed as a share of the population. The model includes county- and state-by-year fixed effects and controls for the interaction between the 1910 population and a post-treatment indicator. Bands report 90% confidence intervals. Standard errors are clustered at the county level.

Figure B.8: Sensitivity Analysis of Confidence Intervals for Non-Linear Pre-Trends



Notes: This figure displays sensitivity analyses of the estimated effect of the pandemic on religiosity (Panel B.8a) and innovation (Panel B.8b) to potential violations of the parallel trends assumption, following Rambachan and Roth (2023). The red bar represents the 90% confidence interval for the first post-pandemic event-study estimate. The blue bars report corresponding 90% robust confidence intervals that allow for per-period violations of parallel trends up to M . Setting M equal to 0 corresponds to a linear differential trend.

B.2 Appendix Tables

Table B.1: Summary Statistics

	(1)	(2)	(3)	(4)	(5)
	Mean	Std. Dev	Min	Max	Counties
Panel A. Mortality					
Flu Excess Deaths	1.127	0.163	0.682	1.827	1275
WWI Deaths	39.834	167.083	0.000	4828.000	1275
Panel B. Religion					
All Denominations - Census Based	40.525	14.099	0.000	94.264	1275
Catholics - Census Based	11.024	12.007	0.000	65.853	1275
Protestants - Census Based	27.448	13.807	0.000	94.264	1275
All Denominations - Name Based	0.016	0.021	-0.039	0.103	1274
Catholics - Name Based	0.001	0.008	-0.030	0.045	1274
Protestants - Name Based	0.014	0.017	-0.027	0.084	1274
Panel C. Scientific Progress					
STEM Employment Share	0.480	0.313	0.054	2.334	1275
Medical Employment Share	0.170	0.093	0.022	1.388	1275
STEM Non-Medical Employment Share	0.310	0.262	0.000	1.912	1275
Total	278.976	1095.265	0.000	15000.000	1275
Pharmaceutical	49.756	200.266	0.000	2708.315	1275
Communication	11.166	54.019	0.000	1110.340	1275
Electrical	40.552	196.034	0.000	4005.376	1275
Mechanical	136.779	533.283	0.000	7834.571	1275
Other	149.124	568.327	0.000	7290.446	1275
Panel D. Income and Demographics					
Population	130.432	329.177	1.061	4819.392	1275
Area	216.975	280.512	0.277	5205.795	1275
Occupational Score per Capita	835.998	74.425	311.939	2302.648	1275
Share of Men	52.305	4.856	19.177	131.582	1275
Share of Illiterates	72.620	9.326	24.633	196.336	1275
Share of Young	33.241	5.981	12.927	86.595	1275
Share of Whites	94.634	11.952	26.163	100.000	1275
Share of African Americans	5.366	11.952	0.000	73.837	1275
Share of Foreign Born	10.680	9.928	0.008	44.052	1275

Notes: This table displays mean, standard deviation, minimum, maximum, and total number of counties for the variables used in the main analysis. Observations are at the county level. Influenza mortality in Panel A is constructed from the mortality statistics; WWI deaths are from Ferrara and Fishback (2024). Data in Panel B are from either the Census of Religious Bodies at the decade level or are constructed from name frequencies at the year level. Patent data in Panel C are from Berkes (2018) and are aggregated at the decade level, while scientific employment is tabulated from the IPUMS full-count 1910 census (Ruggles et al., 2021). The share of STEM is computed from the census and is in 1,000 units. Panel D reports data from the 1910 census. County demographics are measured through the IPUMS full-count census (Ruggles et al., 2021). Panel B data are expressed in percentage terms. Panel D population data are expressed in 1,000 units. All variables are crosswalked to 1920 borders.

Table B.2: Medical, STEM, and Skilled Occupations

Code (1)	Occupation Label (2)	Code (3)	Label (4)
<i>Panel A. Medical Occupations</i>			
19	Medical Sciences	59	Nurses, Student Professional
32	Dentists	75	Physicians and Surgeons
58	Nurses, Professional		
<i>Panel B. STEM Occupations (Includes Medical Occupations)</i>			
7	Chemists	48	Engineering, Mining
12	Agricultural Sciences	49	Engineering (n.e.c.)
13	Biological Sciences	58	Nurses, Professional
14	Chemistry	59	Nurses, Student Professional
16	Engineering	61	Agricultural Scientists
17	Geology and Geophysics	62	Biological Scientists
18	Mathematics	63	Geologists and Geophysicists
19	Medical Sciences	67	Mathematicians
23	Physics	68	Physicists
25	Statistics	69	Miscellaneous Natural Scientists
26	Natural Sciences (n.e.c.)	71	Osteopaths
32	Dentists	75	Physicians and Surgeons
34	Dietitians and Nutritionists	83	Statisticians and Actuaries
41	Engineering, Aeronautical	92	Surveyors
42	Engineering, Chemical	98	Veterinarians
43	Engineering, Civil	240	Officers, Pilots, Purses, and Engineers, Ships
44	Engineering, Electrical	541	Locomotive Engineers
45	Engineering, Industrial	563	Opticians and lens grinders and polishers
46	Engineering, Mechanical	583	Stationery Engineers
47	Engineering, Metallurgical, Metallurgists		
<i>Panel C. Skilled Occupations (Includes STEM and Medical Occupations)</i>			
$1 \leq \cdot \leq 99$	Professional, Technical	$200 \leq \cdot \leq 299$	Managers, Officials, and Proprietors
$500 \leq \cdot \leq 595$	Craftsmen		

Notes: Panel A lists occupations that we classify as “Medical.” Panel B lists occupations that we classify as “Science, Technology, Engineering, and Mathematics” (STEM). Panel C lists occupations that we classify as “Skilled.” Occupation codes and occupation labels are from the IPUMS harmonized 1950 occupation taxonomy (variable “OCC1950”).

Table B.3: Balance Checks: Regressions

	(1)	(2)	(3)
	Coefficient	Standard Error	95% C. I.
<i>Panel A. Religion</i>			
All Denominations (Census Based)	0.057	(0.176)	[-0.245, 0.359]
Catholics (Census Based)	0.168	(0.147)	[-0.084, 0.421]
Protestants (Census Based)	-0.094	(0.149)	[-0.349, 0.161]
All Denominations (Name Based)	0.211	(0.224)	[-0.173, 0.595]
Catholics (Name Based)	-0.079	(0.196)	[-0.415, 0.258]
Protestants (Name Based)	0.350*	(0.191)	[-0.022, 0.678]
<i>Panel B. Scientific Progress</i>			
STEM Employment Share	0.389	(0.281)	[-0.093, 0.870]
Medical Employment Share	0.248	(0.162)	[-0.030, 0.526]
Non-Medical Employment Share	0.377	(0.287)	[-0.115, 0.869]
Total Patents	0.063	(0.055)	[-0.031, 0.158]
Pharmaceutical Patents	0.063	(0.059)	[-0.038, 0.165]
Non-Pharmaceutical Patents	0.063	(0.054)	[-0.030, 0.157]
<i>Panel C. Income and Demographics</i>			
Population Density	-0.009	(0.206)	[-0.363, 0.344]
Share of Men	0.084	(0.056)	[-0.011, 0.179]
Share of Illiterates	-0.147	(0.138)	[-0.384, 0.089]
Share of Young	0.097	(0.146)	[-0.154, 0.348]
<i>Panel D. Ethnic Composition</i>			
Share of Whites	0.096	(0.092)	[-0.062, 0.253]
Share of African Americans	-0.096	(0.092)	[-0.253, 0.062]
Share of Foreign Population	0.310***	(0.110)	[0.122, 0.498]
Immigrants from:			
Italy	0.191	(0.138)	[-0.045, 0.427]
Ireland	-0.141	(0.123)	[-0.351, 0.070]
Austria	0.275**	(0.116)	[0.075, 0.474]
France	-0.044	(0.074)	[-0.171, 0.830]
Spain	-0.007	(0.055)	[-0.101, 0.087]
Portugal	-0.063	(0.157)	[-0.333, 0.207]

Notes: This table displays the correlation between excess deaths defined in equation (3) and a set of covariates in 1910, the last census year before the pandemic. Column (1) reports the standardized coefficient of a regression between the row variable and our measure of excess deaths; column (2) reports the associated standard error in round brackets; column (3) reports the confidence interval of the point estimate at 90% confidence level in square brackets. All variables are expressed as shares of the total population, except for population density. The regressions control for county population and include state-fixed effects. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Impact of the Pandemic on Religiosity: Weighted Regressions

	Share of Affiliated			Name-Based Religiosity		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Catholics	Protestants	All	Catholics	Protestants
Excess Deaths $\times$ Post	0.264** (0.135)	0.119** (0.056)	0.137*** (0.029)	1.052*** (0.274)	0.298 (0.247)	1.084*** (0.229)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	–	–	–
State-Year FE	–	–	–	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	3,825	3,825	3,825	38,220	38,220	38,220
R ²	0.775	0.888	0.925	0.843	0.768	0.844

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on religiosity. The unit of observation is a county observed at a decade frequency between 1906 and 1926 (in columns 1–3) and a yearly frequency between 1900 and 1929 (in columns 4–6). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–3), the dependent variable is the number of individuals affiliated with religious denominations enumerated in the Census of Religious Bodies, divided by county population in 1900; in columns (4–6), the dependent variable is the name-based religiosity measure described in the main text. Columns (1) and (4) report the impact of the pandemic on overall religiosity. Columns (2) and (5) report the impact of the pandemic on Catholicism. Columns (3) and (6) report the impact of the pandemic on Protestantism. All columns include county fixed effects and control for an interaction term between population in 1910 and a post-treatment indicator. Columns (1) to (3) include state-by-decade fixed effects, while columns (4) to (6) include state-by-year fixed effects. Counties are weighted by their 1910 population. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: Impact of the Pandemic on Religiosity: Additional Checks

	(1)	(2)	(3)	(4)
	Name-Based Religiosity			
Excess Deaths Dummy $\times$ Post	0.252*** (0.060)			
Excess Deaths $\times$ Post		0.824*** (0.199)	0.876*** (0.199)	0.001 (0.002)
WWI Deaths $\times$ Post		-0.000 (0.001)		
County FE	Yes	Yes	Yes	Yes
Year–State FE	Yes	Yes	Yes	No
Decade–State FE	–	–	–	Yes
Sample	Baseline	Baseline	No Internal Migrants	Adults
Number of Counties	1,274	1,274	1,274	1,267
Observations	38,220	38,220	38,220	5,068
R ²	0.648	0.648	0.646	0.782

Notes: This table displays the impact of exposure to influenza on overall religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929 in columns (1)–(3) and at decade frequency in column (4). “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918 in columns (1)–(3) and the decades 1920–1930 in column (4)—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). The dependent variable is the name-based measure of aggregate religiosity described in the main text. Column (1) reports the results coding the treatment as a binary variable, returning a value of one if the continuous treatment is above its median and zero otherwise. In column (2), we control for WWI-related deaths interacted with the “Post” indicator. In column (3), we compute religiosity, excluding individuals who were not born in the state where they are recorded in the 1930 census. Column (4) reports results measuring county religiosity using the names stock of adults—which serves as a placebo check. All regressions in columns (1)–(3) include county and state-by-year-fixed effects; the regression in column (4) includes county and state-by-decade-fixed effects. Additionally, each regression controls for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.6: Impact of the Pandemic on Religiosity: Alternative Sample for Name-Based Religiosity

	(1)	(2)	(3)
	Name-Based Religiosity		
Excess Deaths $\times$ Post	1.182*** (0.233)	0.916*** (0.191)	0.123*** (0.026)
County FE	Yes	Yes	Yes
Year–State FE	Yes	Yes	Yes
Sample	No Firstborn	< 5 Kids	Household
Number of Counties	1,274	1,274	1,274
Observations	38,220	38,220	38,220
R ²	0.624	0.631	0.754

Notes: This table displays the impact of exposure to influenza on overall religiosity. The unit of observation is a county, observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). The dependent variable is the name-based measure of aggregate religiosity described in the main text. Column (1) drops first-born children in every household. In column (2), we compute religiosity by dropping all children beyond the fourth in each household. In column (3), we first compute within-household average religiosity and then aggregate the resulting religiosity series at the county-year level. All regressions include county and state-by-year fixed effects. Additionally, each regression controls for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.7: Impact of the Pandemic on Religiosity: Alternative Ways of Constructing Name-Based Religiosity

	Scores w/out County FE		Alternative Thresholds (Unweighted)			Alternative Thresholds (Weighted)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unweighted	Weighted	$(\tau = 1)$	$(\tau = 2)$	$(\tau = 4)$	$(\tau = 1)$	$(\tau = 2)$	$(\tau = 4)$
Excess Deaths $\times$ Post	2.508*** (0.878)	3.572** (1.772)	0.749*** (0.244)	0.852*** (0.238)	0.397** (0.169)	0.901** (0.375)	0.924*** (0.333)	0.390* (0.208)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274	1,274	1,274	1,274	1,274
Observations	38,220	38,220	38,220	38,220	38,220	38,220	38,220	38,220
R ²	0.910	0.977	0.396	0.421	0.692	0.609	0.640	0.855

Notes: This table displays the impact of exposure to influenza on religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1–2), religiosity is measured using religiosity scores obtained by estimating equation (1), except that we do not include the fixed effects in the regression specification. In column (2), counties are weighted by 1900 population. In columns (3–8), religiosity is measured using religiosity scores obtained by estimating equation (1). The term τ denotes the frequency threshold a name must exceed to be included in our sample, in ‰ terms. For instance, $\tau = 2$ implies that at least 2‰ children in our sample must be called with a given name, for that name to be included in the sub-sample of names used to compute the religiosity score. We report the estimated coefficients for different frequency thresholds across the columns. In columns (3–5), counties are not weighted; in columns (6–8), counties are weighted by 1900 population. All specifications include county- and state-by-year fixed effects and control for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.8: Impact of the Pandemic on Religiosity Measured with Saint or Biblical Names

	Religiosity Based on Saints' and Biblical Names		
	(1)	(2)	(3)
	Saints/Biblical	Saints	Biblical
Excess Deaths $\times$ Post	0.316** (0.145)	0.282** (0.136)	0.081** (0.036)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274
Observations	38,220	38,220	38,220
R ²	0.991	0.991	0.985

Notes: This table displays the impact of exposure to influenza on religiosity. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In column (1), the dependent variable is the log of the number of children by cohort whose name either appears in the bible or is the name of a saint; in column (2), the dependent variable only includes biblical names; in column (3), it only includes names of saints. Biblical and saints’ names are from Abramitzky et al. (2016). All specifications include county- and state-by-year fixed effects and control for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.9: Impact of the Pandemic on Religiosity and Innovation: City-Level Analysis

	(1)	(2)
	Name-Based Religiosity	All Patents
Excess Deaths $\times$ Post	0.011*** (0.004)	0.584*** (0.210)
City FE	Yes	Yes
State-Year FE	Yes	Yes
Number of Cities	430	466
Observations	12,900	13,980
R ²	0.639	0.856

Notes: This table displays the city-level effect of exposure to influenza on religiosity (column 1) and innovation measured by patenting activity (column 2). The unit of observation is a city observed at yearly frequency between 1900 and 1929. We report the location of each city in the sample in Figure B.6. The baseline sample is from Clay et al. (2019). We include only cities where we can construct the baseline excess mortality measure. In column (1), the dependent variable is the name-based religiosity measure constructed on the universe of children born between 1900 and 1929 and residing in each city in the 1930 census; in column (2), the dependent variable is $\log(1 + \text{number of patents})$. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). All specifications include city and state-by-year fixed effects. Standard errors, clustered at the city level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.10: Impact of the Pandemic on Employment in Scientific Occupations: Weighted Regressions

	Whole Population			Skilled Population		
	(1)	(2)	(3)	(4)	(5)	(6)
	Share STEM	Share Medical	Share Non-Medical	Share STEM	Share Medical	Share Non-Medical
Excess Deaths $\times$ Post	1.412*** (0.338)	0.393** (0.186)	1.019*** (0.172)	16.508*** (3.425)	4.653*** (1.766)	11.855*** (1.979)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	5,098	5,098	5,098	5,096	5,096	5,096
R ²	0.774	0.770	0.765	0.703	0.715	0.687

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at a decade frequency between 1900 and 1930. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. All specifications include county fixed effects, state-by-year fixed effects, and a control for the interaction between the 1910 population and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. Counties are weighted by their 1910 population. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.11: Impact of the Pandemic on Innovation: Weighted Regressions

	Overall Effect			Intensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
	All Patents	Pharmaceutical	Non-Pharmaceutical	All Patents	Pharmaceutical	Non-Pharmaceutical
Excess Deaths $\times$ Post	0.965*** (0.170)	0.808*** (0.131)	0.869*** (0.165)	1.268*** (0.411)	1.244*** (0.307)	1.149*** (0.403)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	265	265	265
Observations	38,250	38,250	38,250	7,950	7,950	7,950
R ²	0.957	0.932	0.955	0.952	0.914	0.948

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the log(1+ number of granted patents). In columns (2) and (5), the dependent variable is the log(1+ number of granted pharmaceutical patents). In columns (3) and (6), the dependent variable is the log(1+ number of granted non-pharmaceutical patents). In columns (4) to (6), we restrict the sample to counties with at least one granted patent every year. All regressions include county fixed effects, state-by-decade fixed effects, and a control for the interaction between population in 1910 and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. Counties are weighted by their 1910 population. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.12: Impact of the Pandemic on Employment in Scientific Occupations: Excluding Internal Migrants

	Whole Population			Skilled Population		
	(1)	(2)	(3)	(4)	(5)	(6)
	Share STEM	Share Medical	Share Non-Medical	Share STEM	Share Medical	Share Non-Medical
Excess Deaths $\times$ Post	0.383*** (0.049)	0.156*** (0.023)	0.227*** (0.033)	6.722*** (1.855)	2.610*** (0.807)	4.112*** (1.137)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,274	1,274	1,274
Observations	5,098	5,098	5,098	5,096	5,096	5,096
R ²	0.786	0.731	0.786	0.668	0.683	0.628

Notes: This table displays the impact of exposure to the Great Influenza Pandemic on employment in scientific occupations. The unit of observation is a county, observed at a decade frequency between 1900 and 1930. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In columns (1) and (4), the dependent variable is the share of individuals employed in STEM occupations. In columns (2) and (5), the dependent variable is the share of individuals employed in medical occupations. In columns (3) and (6), the dependent variable is the share of individuals employed in a STEM non-medical occupation. In columns (1) to (3), the share is computed with respect to the full population. In columns (4) to (6), we restrict the denominator to include only individuals employed in skilled occupations. We exclude internal migrants from the sample, defined as those born in a different state than where they are recorded in the 1930 census. All specifications include county fixed effects, state-by-year fixed effects, and a control for the interaction between the 1910 population and a post-treatment indicator. Standard errors, clustered at the county level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.13: Impact of the Pandemic on Innovation: Robustness to Alternative Specifications

	All Patents		f (All Patents)		
	(1)	(2)	(3)	(4)	(5)
			Count	$\text{arcsinh}(\cdot)$	Poisson
Excess Deaths Dummy $\times$ Post	0.080***				
	(0.019)				
Excess Deaths $\times$ Post		0.336***	3.736**	0.410***	1.260***
		(0.056)	(1.790)	(0.069)	(0.226)
WW1 Deaths $\times$ Post		3.326			
		(6.144)			
County FE	Yes	Yes	Yes	Yes	Yes
State-Year FE	Yes	Yes	Yes	Yes	Yes
Number of Counties	1,275	1,275	1,275	1,275	1,256
Observations	38,250	38,250	38,250	38,250	37,680
R ²	0.858	0.858	0.911	0.840	0.907

Notes: This table displays the impact of exposure to influenza on innovation. The unit of observation is a county observed at yearly frequency between 1900 and 1929. In columns (1–2), the dependent variable is $\log(1 + \text{total number of patents})$. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). Column (1) reports the results when the treatment is coded as a binary variable equal to one if the continuous variable is above its median and zero otherwise. Columns (2) controls for WW1 deaths interacted with the post-treatment indicator. The dependent variable in column (3) is the raw patent count. In column (4), the dependent variable is the inverse hyperbolic sine of the raw patent count. In column (5), the model is estimated as a Poisson regression, and the dependent variable is the raw patent count. All specifications include county- and state-by-year fixed effects and control for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.14: Impact of the Pandemic on Breakthrough Innovation

	(1)	(2)
	Breakthrough	Share Breakthrough
Excess Deaths $\times$ Post	0.166*** (0.043)	0.016** (0.008)
County FE	Yes	Yes
State-Year FE	Yes	Yes
Number of Counties	1,275	1,275
Observations	38,250	38,250
R ²	0.797	0.265
Mean Dep. Var.	0.336	0.052

Notes: This table displays the impact of the Influenza on patent importance. The unit of observation is a county, observed annually between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). Importance measures are from Kelly et al. (2021). They measure the “importance” of a patent based on the textual similarity between that patent and previous and future patents. In column (1), the dependent variable is the $\log(1 + \text{number of breakthrough patents})$, defined as those in the top 20% of the patent-importance distribution. In column (2), the dependent variable is the $\log(1 + \text{share of breakthrough patents})$. All specifications include county- and state-by-year fixed effects and control for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and displayed in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.15: The Impact of the Pandemic on Religiosity and Innovation

	Pre Pandemic	Post Pandemic	Pooled	DiD
	(1)	(2)	(3)	(4)
Patents per Capita				
Religiosity	-0.059*	0.084**	-0.207***	-0.649***
	(0.035)	(0.041)	(0.042)	(0.241)
Religiosity $\times$ Post			0.659***	
			(0.084)	
Religiosity $\times$ Excess Deaths				0.404*
				(0.211)
Excess Deaths $\times$ Post				0.053***
				(0.014)
Excess Deaths $\times$ Religiosity $\times$ Post				0.556***
				(0.072)
County FE	Yes	Yes	Yes	Yes
State–Year FE	Yes	Yes	Yes	Yes
Number of Counties	1,274	1,274	1,274	1,274
Observations	22,932	15,288	38,220	38,220
R ²	0.548	0.734	0.578	0.580

Notes: This table displays the correlation between innovation and religiosity by exposure to the pandemic. The dependent variable is the number of patents divided by county population in 1910, expressed in 1,000 units, and winsorized at the top and bottom 1% level. The unit of observation is a county, observed annually between 1900 and 1929. “Post” is a categorical variable equal to one during and after the pandemic—i.e., in or after 1918—and zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). In column (1), we display the correlation between religiosity and innovation before the influenza (before 1918); in column (2), we replicate this exercise for the post-pandemic years; in column (3), we pool the years together, and interact religiosity with a post-pandemic indicator; column (4) reports the differential effect of the pandemic by religiosity. All specifications include county- and state-by-year fixed effects and control for the interaction between the 1910 population and a post-treatment indicator. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.16: Impact of the Pandemic on Occupational Choice by Religious Background

	(1)	(2)
	Employed in STEM	Employed in Skilled Occupation
Excess Deaths $\times$ Post $\times$ High Religious Background	-0.701** (0.337)	-0.001 (0.008)
Individual Controls	Yes	Yes
County–Year FE	Yes	Yes
High Religious Background–Year FE	Yes	Yes
High Religious Background–County FE	Yes	Yes
Number of Counties	1,275	1,275
Observations	5,347,751	5,347,751
R ²	0.016	0.048

Notes: This table displays the impact of exposure to the pandemic on occupational choice by individual-level background religiosity. The unit of observation is an adult. In column (1), the dependent variable “Employed in STEM” is an indicator equal to one if an individual is employed in a STEM occupation—as defined in Table B.2—or zero otherwise. In column (2), the dependent variable “Employed in Skilled Occupation” is an indicator equal to one if an individual is employed in a skilled occupation—as defined in Table B.2—or zero otherwise. “Post” is an indicator equal to one if an individual is aged between 18 and 25 in 1918 or zero otherwise. The baseline treatment “Excess Deaths” is defined in equation (3). “High Background Religiosity” is an indicator variable taking the value of one if the religiosity score of the name of the head of the household is equal to or above the median, or zero if it is below the median. The table displays the coefficient of the interaction between these terms. All specifications include a vector of individual controls for the household head: race, urban status, and place of birth, as well as county-by-year, high religious background-by-year, and high religious background-by-county fixed effects. The sample includes the entire population. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.17: The Polarization of Religiosity

	Name-Based Religiosity
Excess Deaths $\times$ Post $\times$ 1st Quintile of Background Religiosity	-0.045** (0.018)
Excess Deaths $\times$ Post $\times$ 2nd Quintile of Background Religiosity	-0.024* (0.014)
Excess Deaths $\times$ Post $\times$ 4th Quintile of Background Religiosity	0.017 (0.015)
Excess Deaths $\times$ Post $\times$ 5th Quintile of Background Religiosity	0.054*** (0.020)
Number of Counties	1,275
Observations	4,246,352
R ²	0.074

Notes: This table displays the regression estimates reported in Figure 6. The unit of observation is a head of household, observed once for each child born between 1900 and 1929 in the household. The observations are weighted by the inverse of the number of children in the household. “Post” is a categorical variable equal to one for children born during and after the pandemic—i.e., in or after 1918—or zero for those born before the pandemic—i.e., before 1918. The baseline treatment “Excess Deaths” is defined in equation (3). Background religiosity is measured as the religiosity score of the child’s father. The third quintile of background religiosity serves as the baseline. The dependent variable “Name-Based Religiosity” is defined as the religiosity score associated with the child’s name. The specification includes fixed effects at the household-head level: race, urban status, and place of birth. It also includes county-by-cohort, county-by-quintiles of background religiosity, and quintiles of background religiosity-by-cohort fixed effects. Standard errors are clustered at the county level and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Summary of Robustness Analyses

Panel A: Religiosity		
Exhibit	Topic	Description
1. Is our name-based measure indeed capturing religiosity?		
i) Table B.6	Accounting for birth order and fertility	One concern is that our results are driven by: i) firstborns, who may be more likely to be named after grandparents (who may have more religious names); ii) numerous families having idiosyncratic naming patterns correlated with religiosity; iii) more religious families having higher fertility. To address these concerns, in Table B.6, we show that our results hold when we drop firstborns (column 1); we drop children beyond the fourth (column 2); we compute household-level average religiosity by assigning to every child a weight that is inversely related to the number of children in the household (column 3).
ii) Table B.7	Religiosity scores without county FE	Our results may be sensitive to how we compute religiosity scores. In Table B.7, columns (1–2) show that our findings hold when we drop county-fixed effects from the measurement equation. Religiosity scores computed this way are identified from the cross-sectional variation in the stock of religiosity across counties, rather than from within-county changes.
iii) Table B.7	Alternative name frequency thresholds	In our main analysis, we impose a threshold of 0.3% for names to be included in the sample for which we estimate religiosity scores. Our results are robust when using alternative frequency thresholds.
iv) Table B.8	Alternative religiosity measures	In Table B.8, we compute a religiosity indicator of first names based on the names of biblical figures and saints collected by Abramitzky et al. (2016). Our results are robust to using this measure.
v) Figure B.1	Predicted vs. measured religiosity	We find a positive correlation between predicted religiosity and religiosity reported in the Census of Religious Bodies.

vi)	Figure B.4	Saint or biblical names	Our measure of religiosity is highly and positively correlated with the one developed by Abramitzky et al. (2016).
-----	------------	-------------------------	--

2. Was the increase in religiosity caused by WW1?

i)	Table B.5, Figure B.3	WW1 and religiosity	In column (2), we interact WW1 deaths with a post-pandemic indicator. The estimated coefficient is not significant. Additionally, in Figure B.3, we show that WW1 deaths and our measure of excess deaths are not correlated.
----	--------------------------	---------------------	---

3. Was the increase in religiosity driven by migrations?

i)	Table B.5	Internal migrants	We check that the effect of the influenza on religiosity is not driven by individuals who migrated to their residence in 1930 from another state, perhaps responding to a slacker labor market following the pandemic (column 3).
ii)	Table B.5	Religiosity of adults	Our results may be driven by selective migrations of more religious individuals towards areas more affected by the pandemic. In column (4), we use the average religiosity score of adults' own names (a measure of an individual's background religiosity) as the dependent variable. As this variable was determined before the pandemic, a significant effect of the Influenza would suggest internal or international migrations correlated with background religiosity and the intensity of the shock. We do not find evidence in support of this mechanism.

4. Is the observation unit (counties) driving the results?

i)	Table B.9	Religiosity of cities	The baseline analysis is performed at the level of counties. We estimate the effect of the influenza on religiosity at the city level using the sample of Clay et al. (2019). We replicate the main results in column 1.
----	-----------	-----------------------	--

5. Are the results driven by particular states?

i)	Figure B.5a	Exclude one state at a time	We exclude one state at a time from the estimation sample and show that the estimated treatment effect remains stable irrespective of the excluded state.
----	-------------	-----------------------------	---

6. Are our results sensible to violations of parallel trends?

i)	Figure B.8a	Parallel trends violations	We employ the methodology developed by Rambachan and Roth (2023) and find that the estimated treatment effect remains robust when we allow for substantial deviations from the baseline parallel trends assumption.
----	-------------	----------------------------	---

Panel B: Innovation

Exhibit	Topic	Description
7. Is our measure indeed capturing innovation activity?		
i)	Table B.13	Alternative measures of innovation
		In columns (3) to (5), we use alternative transformations of the raw patent count (explained in the header of Tables B.13). The results are quantitatively stable across specifications.
8. Was the increase in innovation caused by WW1?		
i)	Table B.13, Figure B.3	WW1 and innovation
		In column (2), we interact WW1 deaths with a post-pandemic indicator. The estimated coefficient is not significant. Additionally, in Figure B.3, we show that WW1 and influenza-related deaths are uncorrelated.
9. Did the pandemic trigger an increase in high- (or low-) impact innovation?		
i)	Table B.14	Breakthrough innovation
		Using the measure defined by Kelly et al. (2021), we show that the number of “breakthrough” patents, <i>i.e.</i> those in the upper 20% of the patent impact distribution, as well as its share with respect to all patents, increases in counties hit harder by the pandemic.
10. Is the observation unit (counties) driving the results?		

i)	Table B.9	Innovation of cities	The baseline analysis is performed at the county level. We estimate the effect of the influenza on innovation at the city level using the sample of Clay et al. (2019). We replicate the main results in column (2).
----	-----------	----------------------	--

11. Was the increase in STEM driven by migrations?

i)	Table B.12	Excluding internal migrants	One results may be driven by selective migrations of STEM individuals towards areas more affected by the pandemic. We explicitly test this by excluding individuals who were born in a different state than the one where they were recorded in the 1930 census. Results remain under both sample restrictions.
----	------------	-----------------------------	---

12. Are our results driven by particular states?

i)	Figure B.5b	Exclude one state at a time	We exclude one state at a time from the estimation sample and show that the estimated treatment effect remains stable irrespective of the excluded state.
----	-------------	-----------------------------	---

12. Are our results sensitive to pre-treatment differences in observables (e.g., unemployment rate)?

i)	Figure B.7	Heterogeneous Responses	We add each control individually to the baseline specification, measuring it at its 1910 value (the last census before the pandemic) as a share of the 1910 population and interacting it with the post-influenza indicator. The coefficients remain virtually unchanged.
----	------------	-------------------------	---

13. Are our results sensible to violations of parallel trends?

i)	Figure B.8b	Parallel trends violations	We employ the methodology developed by Rambachan and Roth (2023) and find that the estimated treatment effect remains robust when we allow for substantial deviations from the baseline parallel trends assumption.
----	-------------	----------------------------	---
